

A delta-tubulin/epsilon-tubulin/Ted protein complex is required for centriole architecture

Reviewed Preprint

v2 • February 6, 2025

Revised by authors

Reviewed Preprint

v1 • July 2, 2024

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The study by Pudlowski et al. shows that a previously-identified protein complex, composed of delta- and epsilon-tubulin together with TEDC1 and TEDC2, functions in generating centriolar triplet microtubules, and that this is crucial for the proper formation of centriolar subdomains and the stability of centrioles throughout the cell cycle. This is an **important** study that advances our understanding of centriole biogenesis and structure and is supported by **convincing** evidence based on knockout cell lines, immunoprecipitation, and ultrastructure expansion microscopy. The work is of interest to cell biologists, in particular researchers with interest in centrosome biology.

<https://doi.org/10.7554/eLife.98704.2.sa4>

Abstract

Centrioles have a unique, conserved architecture formed by three linked “triplet” microtubules arranged in nine-fold symmetry. The mechanisms by which these triplet microtubules are formed are not understood and likely involve the noncanonical tubulins delta-tubulin and epsilon-tubulin. Previously, we found that human cells deficient in delta-tubulin or epsilon-tubulin form abnormal centrioles, characterized by an absence of triplet microtubules, lack of central core protein POC5, and a futile cycle of centriole formation and disintegration (Wang et al., 2017). Here, we show that human cells lacking either of the associated proteins TEDC1 and TEDC2 have these same phenotypes. Using ultrastructure expansion microscopy, we find that mutant centrioles elongate to the same length as control centrioles in G2-phase. These mutants fail to recruit inner scaffold proteins of the central core and have an expanded proximal region. During mitosis, the mutant centrioles elongate further before fragmenting and disintegrating. All four proteins physically interact and TEDC1 and TEDC2 are capable forming a subcomplex in the absence of the tubulins. These results support an AlphaFold Multimer model of the tetramer in which delta-tubulin and epsilon-tubulin are predicted to form a heterodimer. TEDC1 and TEDC2 localize to centrosomes and are mutually dependent on each other and on delta-tubulin and epsilon-tubulin for localization. Our results demonstrate that delta-tubulin, epsilon-tubulin, TEDC1,

and TEDC2 function together to promote robust centriole architecture. This work also lays the groundwork for future molecular studies of this complex, providing a basis for determining the mechanisms that underlie the assembly and interplay between the triplet microtubules and inner centriole structure.

Introduction

The major microtubule organizing center of mammalian cells, the centrosome, is composed of two barrel-shaped centrioles surrounded by layers of pericentriolar material (Breslow and Holland, 2019). The unique architecture of the centriole is highly conserved: the centriole barrel walls of approximately 250 nm in diameter by 500 nm in length are formed of compound microtubules linked to each other through shared protofilament walls, arranged in nine-fold symmetry (Wang and Stearns, 2017). Centrioles exhibit proximal-distal polarity comprised of three subdomains: the proximal end with triplet microtubules, the distal end with doublet microtubules, and the central core spanning the two regions (LeGuennec et al., 2021). The triplet microtubules are named the A-, B-, and C-tubules. The A-tubule is a complete microtubule formed of 13 protofilaments, and the B- and C-tubules are partial tubules and share protofilament walls with adjacent tubules. The A- and B-tubules extend beyond the C-tubule to form the doublet microtubules of the centriole distal end. During ciliogenesis, the A- and B-tubules elongate further to form the ciliary axoneme (Wang and Stearns, 2017).

Compound microtubules are unique to centrioles and ciliary axonemes and are conserved in almost all organisms with these organelles. Little is known about the mechanisms by which they form, or the functional roles they play within centrioles and cilia. Two non-canonical members of the tubulin superfamily, delta-tubulin (TUBD1) and epsilon-tubulin (TUBE1), are required for compound microtubule formation or stability in multiple organisms (de Loubresse et al., 2001; Dupuis-Williams et al., 2002; Dutcher and Trabuco, 1998; Dutcher et al., 2002; Gadelha et al., 2006; Goodenough and StClair, 1975; Ross et al., 2013; Wang et al., 2017). Previously, we showed that human cells lacking these tubulins make aberrant centrioles that only have singlet microtubules and disintegrate in mitosis, resulting in a futile cycle of centriole formation and loss every cell cycle (Wang et al., 2017). These mutant centrioles fail to recruit the distal end protein POC5, indicating that compound microtubules may be required for centriole composition. We concluded that either the compound microtubules themselves, or the proteins that they associate with, are required for centriole stability through the cell cycle. Together, these results suggest that the compound microtubules may form a unique scaffold for the protein-protein interactions that define centrosomes and cilia.

The compound microtubules are directly linked to many of the substructures at the proximal, central, and distal regions within centrioles. At the proximal end, the cartwheel, a ninefold symmetric hub and spokes made from SASS6 and associated proteins, is connected to the A-tubule through the pinhead, which has been proposed to be formed of CEP135 and CPAP (Hatzopoulos et al., 2013; Kraatz et al., 2016; Lin et al., 2013a; Sharma et al., 2016). Multiple cartwheels are stacked within the centriole lumen to a height of approximately one-third of the entire centriole length (~170 nm in human centrioles) (Klena et al., 2020). The A-tubule of one triplet is connected to the C-tubule of the adjacent triplet through a structure known as the A-C linker. Recently CCDC77, WDR67, and MIIP were identified to be components of the A-C linkers (Bournonville et al., 2024; Laporte et al., 2024). Within the central core, a helical inner scaffold imparts structural integrity upon the centriole (Le Guennec et al., 2020; Steib et al., 2020), and recruits proteins, including gamma-tubulin, to the lumen of the centriole (Schweizer et al., 2021). This scaffold is formed in G2-phase of the first cell cycle after centriole birth, is composed of POC5, POC1B, FAM161A, WDR90, and CCDC15 and contacts all three (A-, B-, and C-) tubules of the triplet (Arslanhan et al., 2023; Laporte et al., 2024; Le Guennec et al., 2020; Steib et al., 2020). The distal region of centrioles also has a unique protein composition, including the

proteins centrin, CP110, SFI1, CEP97, CEP90, OFD1, and MNR (Kleylein-Sohn et al., 2007 [↗](#); Kumar et al., 2021 [↗](#); Laporte et al., 2022 [↗](#); Laporte et al., 2024 [↗](#); Le Borgne et al., 2022 [↗](#); Spektor et al., 2007 [↗](#)). The connections between the compound microtubules and these distal end proteins are not well-understood.

Canonically, centriole formation in cycling cells is “templated,” in which one newly formed procentriole is created at the proximal end of each pre-existing parental centriole in S-phase, resulting in four centrioles within the cell. During the first cycle after their formation, procentrioles acquire post-translational modifications, elongate, recruit the inner scaffold, lose the cartwheel, and undergo centriole-to-centrosome conversion. Additional changes occur during the second cell cycle, including acquisition of the distal and subdistal appendages that are important for ciliogenesis (Sullenberger et al., 2020 [↗](#); Tischer et al., 2021 [↗](#)). Under experimental manipulations in which the parental centrioles are ablated, centrioles can also form *de novo* in S-phase. *De novo* centriole formation can result in more than five centrioles per cell and has been shown to be error-prone (Wang et al., 2015 [↗](#)), perhaps indicating differences in centriole structure or regulation. The composition and architecture of centrioles made in this manner has not been systematically characterized.

Here, we extend our original work by defining the roles of two additional proteins, TEDC1 and TEDC2, that regulate triplet microtubule formation and stability. These proteins physically interact with TUBD1 and TUBE1 (Breslow et al., 2018 [↗](#); Huttlin et al., 2017 [↗](#); Huttlin et al., 2021 [↗](#)). Loss of *Tedc1* or *Tedc2* in 3T3 cells results in a variable distribution of centriole numbers through the cell cycle, and tagged TEDC1 localizes to centrosomes (Breslow et al., 2018 [↗](#)). We created *TEDC1*^{-/-} or *TEDC2*^{-/-} mutant cells in the same background as the *TUBD1*^{-/-} and *TUBE1*^{-/-} mutants and found that these cells phenocopy loss of TUBD1 or TUBE1. All four proteins interact in a complex. We find that the compound microtubules are required for recruiting the helical inner scaffold and correctly positioning the proximal end. As part of our analysis, we also determine the composition and architecture of centrioles formed *de novo* and find that these are very similar to those of procentrioles formed by templated centriole duplication. Together, these results indicate that compound microtubules are required for scaffolding substructures within centrioles and maintaining centriole stability through the cell cycle.

Results

Loss of TEDC1 or TEDC2 phenocopies loss of TUBD1 or TUBE1

TEDC1 and TEDC2 have been reported to physically interact with delta-tubulin and epsilon-tubulin, and loss of either *Tedc1* or *Tedc2* in 3T3 cells results in cells with a variable number of centrioles through the cell cycle (Breslow et al., 2018 [↗](#)). To further dissect the phenotypes of loss of *TEDC1* or *TEDC2* and directly compare to our original report on delta-tubulin and epsilon-tubulin, we used CRISPR/Cas9 to generate strong loss of function/null mutations in *TEDC1* or *TEDC2* in the same cell type and background genotype (hTERT RPE-1 *TP53*^{-/-}, which will be referred to as RPE-1 *p53*^{-/-}) as the *TUBD1*^{-/-} (delta-tubulin knockout) and *TUBE1*^{-/-} (epsilon-tubulin knockout) mutant cells (Fig 1 - Supp 1 [↗](#)). By immunofluorescence staining for two centriolar proteins, centrin (CETN) and CP110, we observed that *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant cells had similar phenotypes to each other and to *TUBD1*^{-/-} and *TUBE1*^{-/-} mutant cells: in an asynchronously growing culture, about half of the cells had no centrioles, and half had five or more centrioles. These phenotypes were fully rescued by expression of tagged TEDC1 (TEDC1-Halotag-3xFlag) or TEDC2 (TEDC2-V5-APEX2) (Fig 1A [↗](#), Fig 1 - Supp 1 [↗](#)).

Next, we checked whether the centrioles in *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant cells underwent a futile cycle of centriole formation and disintegration. We synchronized cells in each stage of the cell cycle, quantified the number of cells with centrioles, and found that almost all mutant cells lacked

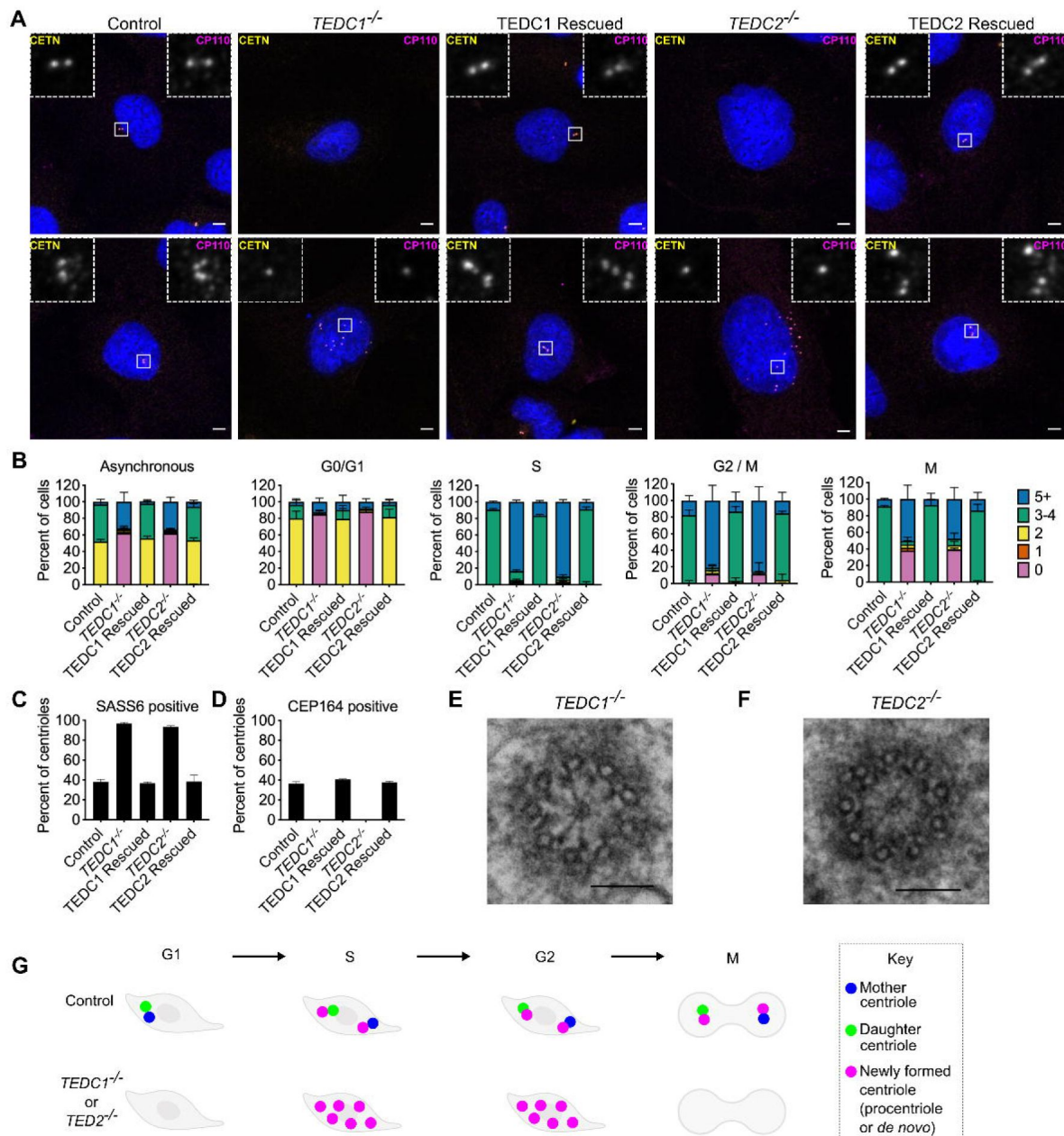


Figure 1.

Loss of TEDC1 or TEDC2 phenocopies loss of delta-tubulin or epsilon-tubulin

(A) Immunofluorescence staining of control (RPE1 *TP53*^{-/-}), *TEDC1*^{-/-} (RPE1 *TP53*^{-/-}; *TEDC1*^{-/-}), TEDC1 Rescued (RPE1 *TP53*^{-/-}; *TEDC1*^{-/-}; TEDC1-Halotag-3xflag), *TEDC2*^{-/-} (RPE1 *TP53*^{-/-}; *TEDC2*^{-/-}), TEDC2 Rescued (RPE1 *TP53*^{-/-}; *TEDC2*^{-/-}; Tedc2-V5-APEX2) cells. Top row: G1 stage cells with 2 centrioles. Bottom row: S/G2 stage cells with 4 centrioles. Blue: DAPI; Yellow: Centrin (CETN); Magenta: CP110. Images are maximum projections of confocal stacks. Scale bar: 5 μ m

(B) Centriole number counts of the indicated cell lines. Cells were either asynchronous, serum-starved for G0/G1, stained for PCNA for S-phase, synchronized with RO-3306 for G2/M, or mitotic figures were identified by DAPI staining. Each condition was performed in triplicate, with n=100 cells scored for each.

(C) Percent of all centrioles (parental, pro, and *de novo* centrioles) in indicated cell types positive for SASS6 staining. Each condition was performed in triplicate, with 200 cells scored for each.

(D) Percent of all centrioles (parental, pro, and *de novo* centrioles) in indicated cell types positive for CEP164 staining. Each condition was performed in triplicate, with 100 cells scored for each.

(E) TEM cross-section of a centriole in a G2-phase *TEDC1*^{-/-} cell. Scale bar: 100 nm

(F) TEM cross-section of a centriole in a G2-phase *TEDC2*^{-/-} cell. Scale bar: 100 nm

(G) Schematic of centriole formation and loss in control and *TEDC1*^{-/-} or *TEDC2*^{-/-} cells.

centrioles in G0/G1 phase. Centrioles formed in S-phase and disintegrated in M (**Fig 1B**). The centrioles that were present in mutant cells were immature: all centrioles were positive for the procentriole marker SASS6 and negative for the mature centriole marker CEP164 (**Fig 1C**, **1D**). We conclude that cells lacking TEDC1 or TEDC2 also undergo a futile cycle, similar to cells lacking delta-tubulin or epsilon-tubulin (**Fig 1G**).

We also examined the centriolar microtubule status of *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant cells by TEM. Similar to cells lacking delta-tubulin or epsilon-tubulin, we found that centrioles in *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant cells lacked compound microtubules and only had singlet microtubules. These centrioles had cartwheels and pinheads, but A-C linkers were not visible (**Fig 1E,F**, **Fig 1 – Supp 2**). Together, these results demonstrate that loss of TEDC1 or TEDC2 phenocopies loss of delta-tubulin or epsilon-tubulin, indicating that these proteins likely act together.

TEDC1 and TEDC2 localize to centrosomes

Next, we investigated the localization of TEDC1 and TEDC2 to determine if they may directly act on centrosomes. TEDC1 and TEDC2 are expressed at low levels in cells (**Fig 1 – Supp 1**), and we could not reproducibly localize the endogenous proteins with antibody staining. Instead, we localized the functional, tagged proteins in our rescue cell lines. We found that the tagged rescue constructs localize to centrosomes, (**Fig 2A** and **2B**) and the antibodies for the tags were specific (**Fig 2 – Supp Fig 1E-J**). TEDC1 and TEDC2 were enriched at centrosomes in S/G2 and colocalized with SASS6, but not centrin, indicating that TEDC1 and TEDC2 may localize to newly formed procentrioles and/or the proximal ends of parental centrioles.

To analyze TEDC1 and TEDC2 localization at higher resolution, we localized our tagged rescue constructs using three methods: a super-resolution spinning disk confocal microscope with immunofluorescence microscopy (**Fig 2 – Supp Fig 1A,B**), ultrastructure expansion microscopy (U-ExM, (Gambaretto et al., 2019), **Fig 2C, D**), and a second expansion microscopy method (Kong et al., 2024, **Fig 2 – Supp 1C, D**). With all three methods, we observed that both proteins localize to procentrioles and the proximal ends of parental centrioles. At these regions, both proteins overlap with the centriolar microtubules. Together, these results show that TEDC1 and TEDC2 localize to centrosomes and likely directly act upon them.

TEDC1, TEDC2, TUBD1 and TUBE1 form a complex in cells

To determine how TEDC1, TEDC2, TUBD1 and TUBE1 might act together, we first determined whether they are mutually required for their localization at centrosomes. We found that TEDC1 did not localize to centrioles in the absence of TEDC2, TUBD1, or TUBE1 (**Figure 3A**). Likewise, TEDC2 did not localize to centrioles in the absence of TEDC1, TUBD1, or TUBE1 (**Figure 3B**). These results indicate that these proteins are mutually required for TEDC1 or TEDC2 localization. Furthermore, overexpression of TEDC1 or TEDC2 did not rescue the centriole phenotypes in any of the other mutants, indicating that TEDC1 and TEDC2 are not downstream effectors of TUBD1 and TUBE1 (**Fig 3A** and **3B**).

TEDC1 and TEDC2 have previously been shown to physically interact with TUBD1 and TUBE1 (Breslow et al., 2018). To further probe the nature of this interaction, we first determined whether any of these proteins may form subcomplexes in cells. We expressed TEDC1-HaloTag-3xFlag in each mutant cell line and determined whether immunoprecipitation of tagged TEDC1 could precipitate the other proteins. TEDC1-HaloTag-3xFlag rescuing the *TEDC1*^{-/-} mutant could precipitate TEDC2, TUBD1, and TUBE1, indicating that all four proteins physically interact. TEDC1 did not interact with epsilon-tubulin in the absence of delta-tubulin, nor did it interact with delta-tubulin in the absence of TUBE1. In the absence of TEDC2, TEDC1 did not interact with TUBD1 or TUBE1. However, in the absence of TUBD1 or TUBE1, TEDC1 and TEDC2 could still interact with each other (**Fig 3C**).

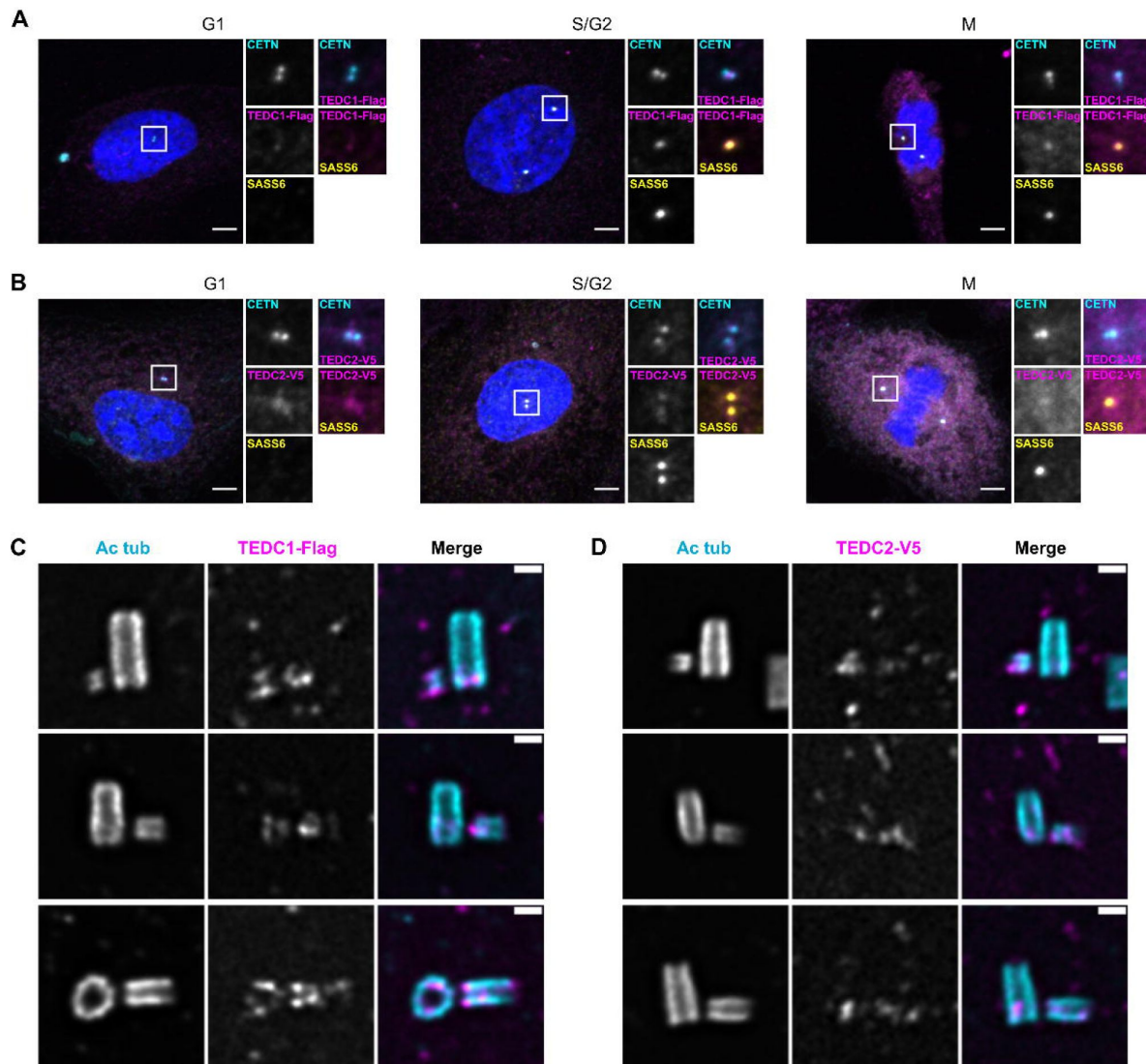


Figure 2.

Tedc1 and Tedc2 localize to centrioles

(A) Immunofluorescence staining of Tedc1 rescue cell lines expressing TEDC1-Halotag-3xFlag in G1, S/G2, and M. Images are maximum projections of confocal stacks. Blue: DAPI; Cyan: Centrin; Magenta: TEDC1-Halotag-3xFlag (localized with anti-Flag antibody); Yellow: SASS6. Scale bar: 5 μ m. **(B)** Immunofluorescence staining of Tedc2 rescue cell lines expressing TEDC2-V5-APEX2 in G1, S/G2, and M. Images are maximum projections of confocal stacks. Blue: DAPI; Cyan: Centrin (localized with anti-GFP antibody recognizing GFP-centrin); Magenta: TEDC2-V5-APEX2 (localized with anti-V5 antibody); Yellow: SASS6. Scale bar: 5 μ m. **(C)** U-ExM of Tedc1 rescue cell lines expressing TEDC1-Halotag-3xFlag, arranged by procentriole length. Cyan: Acetylated tubulin; Magenta: TEDC1-Halotag-3xFlag (localized with anti-Flag antibody). Confocal image stacks were deconvolved using Microvolution; single plane images shown. Scale bar: 1 μ m. **(D)** U-ExM of Tedc2 rescue cell lines expressing TEDC2-V5-APEX2, arranged by procentriole length. Cyan: Acetylated tubulin; Magenta: TEDC2-V5-APEX2 (localized with anti-V5 antibody). Confocal image stacks were acquired with a Yokogawa CSU-W1 spinning disk microscope and deconvolved using Microvolution; single plane images shown. Scale bar: 1 μ m.

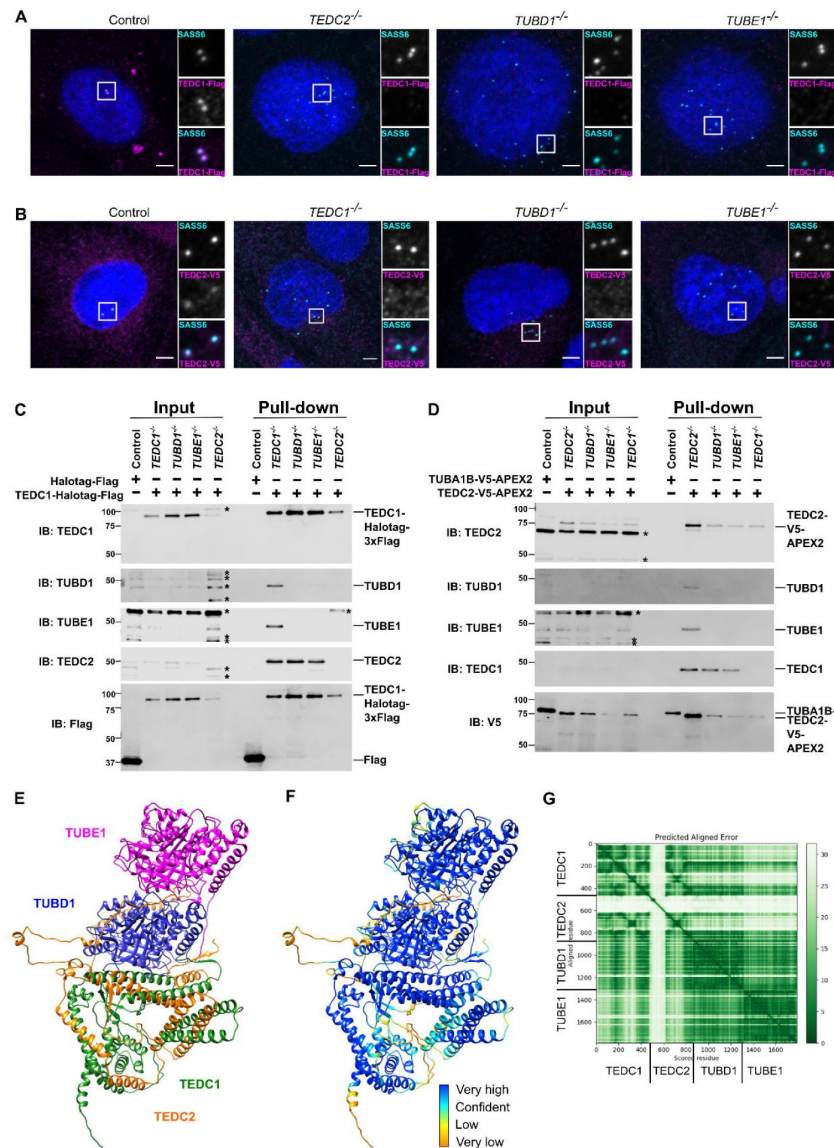


Figure 3.

TEDC1, TEDC2, TUBD1, TUBE1 form a complex in cells (A) Centrosomal TEDC1 localization depends on TEDC2, TUBD1, TUBE1.

Immunofluorescence staining of cells expressing TEDC1-Halotag-3xflag. Control cell is *TEDC1*^{-/-} mutant cells rescued with TEDC1-Halotag-3xflag. Images are maximum projections of confocal stacks. Blue: DAPI; Cyan: SASS6; Magenta: TEDC1-Halotag-3xflag (localized with anti-Flag antibody). Scale bar: 5 μm. **(B) Centrosomal TEDC2 localization depends on TEDC1, TUBD1, TUBE1.** Immunofluorescence staining of cells expressing TEDC2-V5-APEX2. Control cell is TEDC2 mutant cells rescued with TEDC2-V5-APEX2. Images are maximum projections of confocal stacks. Blue: DAPI; Cyan: SASS6; Magenta: TEDC2-V5-APEX2 (localized with anti-V5 antibody). Scale bar: 5 μm. **(C) TEDC1 pulls down TEDC2 in the absence of delta or epsilon-tubulin.** Western blot of input and pull-down of Halotag-Flag or TEDC2-Halotag-Flag in the indicated cell lines. IB: indicates the antibody used for immunoblotting. The proteins and their positions are labeled on the right. Asterisks mark non-specific bands. **(D) TEDC2 pulls down TEDC1 in the absence of delta or epsilon-tubulin.** Western blot of input and pull-down of TUBA1B-V5-APEX2 or TEDC2-V5-APEX2 in the indicated cell lines. IB: indicates the antibody used for immunoblotting. The proteins and their positions are labeled on the right. Asterisks mark non-specific bands. **(E)** AlphaFold-Multimer prediction of the complex **(F)** AlphaFold-Multimer prediction colored according to pLDDT. Very high: pLDDT > 90. High: 90 > pLDDT > 70. Low: 70 > pLDDT > 50. Very low: pLDDT < 50 **(G)** Predicted align error of the AlphaFold Multimer prediction. Expected position error (Angstroms) is graphed.

We performed the reciprocal experiment, in which we expressed TEDC2-V5-APEX2 in each mutant cell line and determined whether immunoprecipitation of tagged TEDC2 could precipitate the other proteins. We observed similar results as our analysis with TEDC1. TEDC2-V5-APEX2 rescuing the *TEDC2*^{-/-} mutant could precipitate TEDC1, TUBD1, and TUBE1, indicating that all four proteins physically interact. TEDC2 did not interact with either tubulin in the absence of the other. In the absence of TEDC1, TEDC2 did not interact with either tubulin. However, in the absence of TUBD1 or TUBE1, TEDC2 and TEDC1 could still interact (**Fig 3D** [↗](#)).

Together, these experiments indicate that TEDC1, TEDC2, TUBD1 and TUBE1 physically interact with each other, as previously reported ([Breslow et al., 2018](#) [↗](#); [Huttlin et al., 2017](#) [↗](#); [Huttlin et al., 2021](#) [↗](#)). Furthermore, TEDC1 and TEDC2 can form a subcomplex in the absence of either tubulin.

To gain additional insight into the nature of this interaction, we used AlphaFold-Multimer ([Evans et al., 2021](#) [↗](#)) to predict the structure of the complex. AlphaFold-Multimer predicted that TUBD1 and TUBE1 would form a heterodimer, similar to the alpha-tubulin/beta-tubulin heterodimer, with TUBD1 at the minus-end of the heterodimer. AlphaFold also predicted that the alpha-helices of TEDC1 and TEDC2 interact with each other, and that TEDC1 and TEDC2 form an interaction surface with TUBD1. These predictions, especially at the interface between TEDC1, TEDC2, and TUBD1, yielded high confidence pLDDT and PAE scores (**Fig 3E-G** [↗](#), **Fig 3 - Supp 1A** [↗](#)). A similar prediction was obtained with the newly released AlphaFold 3 ([Abramson et al., 2024](#) [↗](#)) (**Fig 3 - Supp 1B** [↗](#)). As controls, we used AlphaFold-Multimer to predict whether TEDC1 and TEDC2 might interact with alpha-tubulin and beta-tubulin, and whether similar structures would be predicted for *Xenopus* TEDC1, TEDC2, TUBD1 and TUBE1. While AlphaFold-Multimer did not predict a high-confidence interaction for TEDC1, TEDC2, alpha- and beta-tubulin (**Fig 3 - Supp 1C** [↗](#)), it did predict a high-confidence structure for *Xenopus* TEDC1, TEDC2, TUBD1 and TUBE1, similar to that predicted for the human proteins (**Fig 3 - Supp 1D** [↗](#)).

Our pulldown experiments showed that TEDC1 and TEDC2 can interact in a subcomplex in the absence of TUBD1 or TUBE1, which supports the predicted structural model, in which TEDC1 and TEDC2 are predicted to directly interact with each other without being bridged by either tubulin. Further supporting this model, immunoprecipitation of TEDC2 identifies the other proteins in stoichiometric amounts ([Breslow et al., 2018](#) [↗](#)), and we previously showed that TUBD1 and TUBE1 physically interact ([Wang et al., 2017](#) [↗](#)). Given the size and shape of the tetrameric complex as predicted by AlphaFold-Multimer, it is possible that these may form a structural component of centrioles. Future work will be necessary to test these possibilities. Together, our experiments indicate that TEDC1, TEDC2, TUBD1 and TUBE1 physically interact in a complex and are recruited together to centrioles.

Loss of TEDC1, TEDC2, TUBD1 or TUBE1 results in centrioles with aberrant ultrastructure

Next, we determined how the loss of these proteins, and the triplet microtubules themselves, affect centriole ultrastructure and protein composition. Because centrioles are constitutively formed *de novo* every cell cycle in our mutant cells, we incorporated two controls in our analysis: procentrioles undergoing normal parental-mediated centriole duplication in control (RPE-1 *p53*^{-/-}) cells, and centrioles formed in RPE-1 *p53*^{-/-} cells *de novo* in the first cell cycle after centrinone washout. For each of the 2 control and 4 mutant cell lines, cells were synchronized by mitotic shake off, resulting in coverslips enriched for cells in late S and G2 phases, with a minor population in M phase. Synchronized cells were then expanded using U-ExM and stained for centriolar markers.

We first tested whether the microtubules of mutant centrioles could be modified by acetylation of alpha-tubulin. During centriole formation, acetylation is thought to proceed from the proximal toward the distal end and from the A- to the C-tubules ([Sahabandu et al., 2019](#) [↗](#)). We found that

antibodies against acetylated alpha-tubulin stained mutant centrioles well (**Fig 4B** [↗](#)), indicating that centrioles with only singlet A-tubules can be acetylated.

We next tested whether mutant centrioles were capable of elongating during the cell cycle. In our expansion gels of cells enriched in late S and G2 phases, we used PCNA to mark S-phase cells and co-stained with acetylated tubulin to mark centrioles. Similar to a recently published report, we also found a range of centriole lengths in S- and G2-phases ([Laporte et al., 2024](#) [↗](#)). In S-phase, centrioles were short in all conditions. In G2-phase, centrioles elongated in all conditions, and mutant centrioles reached approximately similar lengths as control centrioles (**Fig 4A** [↗](#)). By contrast, mutant centriole widths did not increase and centrioles remained narrow, as we previously reported (**Fig 4 – Supp 5** [↗](#) and [Wang et al., 2017](#) [↗](#)). These results indicate that centrioles with singlet microtubules can elongate to the same overall length as control centrioles in G2 phase. Consistent with this hypothesis, CEP120, a protein involved in regulating centriole length ([Comartin et al., 2013](#) [↗](#); [Lin et al., 2013b](#) [↗](#); [Mahjoub et al., 2010](#) [↗](#)), was present and properly localized within mutant centrioles (**Fig 4 – Supp 1D** [↗](#)).

The compound microtubules of centrioles are heavily post-translationally modified, and recent studies have indicated that each tubule may acquire different modifications ([Guichard et al., 2023](#) [↗](#)). We checked glutamylation, a post-translational modification thought to be restricted to the outer surface of centrioles ([Guichard et al., 2023](#) [↗](#)). Within *Chlamydomonas* centrioles, glutamylation is differentially distributed between each tubule: on the C-tubule at the distal end, on all 3 tubules in the central core, and on the A-tubule at the proximal end ([Hamel et al., 2017](#) [↗](#)). In human centrioles, polyglutamylation is enriched in the proximal and central regions, and is absent in the distal region ([Gambarotto et al., 2019](#) [↗](#); [Mahecic et al., 2020](#) [↗](#); [Sullenberger et al., 2020](#) [↗](#)). We used two antibodies to detect glutamylation: the GT335 antibody, which recognizes the glutamylation branch and thus detects all polyglutamylation, and the polyE antibody, which recognizes long polyglutamate side chains with at least 2 or 3 glutamate residues ([Kann et al., 2003](#) [↗](#); [Van Dijk et al., 2007](#) [↗](#)). We found that mutant and control centrioles could be stained by GT335 (**Fig 4C** [↗](#)), indicating that mutant centrioles are at least mono-glutamylated. However, the polyE antibody did not label control procentrioles or *de novo* centrioles in the first cell cycle after their formation, making this antibody uninformative for our mutants (**Fig 4D** [↗](#)). These results show that centrioles with just singlet microtubules (A-tubules) can be mono-glutamylated. Moreover, similar to previous reports ([Sullenberger et al., 2020](#) [↗](#)), our results suggest that centriole glutamylation is a multi-step process, in which long glutamate side chains are added later during centriole maturation.

We previously demonstrated that *TUBD1*^{-/-} and *TUBE1*^{-/-} mutant centrioles fail to recruit the distal centriole protein POC5 ([Wang et al., 2017](#) [↗](#)). Using expansion microscopy, we found that *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant centrioles also failed to recruit POC5 (**Fig 4Ei** [↗](#)). Since our original work was published, POC5 was shown to be a component of the helical inner scaffold within the central core. These results indicate that the helical inner scaffold is not properly formed in centrioles with singlet microtubules. To test the mechanisms underlying loss of POC5, we next tested whether mutant centrioles recruit WDR90, which has been proposed to localize to the inner junction between the A- and B-tubules and function in recruiting the inner scaffold ([Steib et al., 2020](#) [↗](#)). We found that WDR90 was not recruited to mutant centrioles, in contrast to control centrioles, in which it is recruited in G2-phase (**Fig 4Eii** [↗](#)). From these results, it is likely that mutant centrioles with singlet microtubules fail to build or stabilize the inner junction between the A- and B-tubules. In the absence of the inner junction and junctional protein WDR90, centrioles with singlet microtubules cannot form the inner scaffold. As also previously reported ([Laporte et al., 2024](#) [↗](#)), we failed to detect gamma-tubulin within the lumen of control or *de novo*-formed centrioles in S or G2-phase (**Fig 4-Supp1E** [↗](#)) and thus were unable to test whether gamma-tubulin, which is recruited to the lumen of centrioles by the inner scaffold, was mislocalized in mutant centrioles.

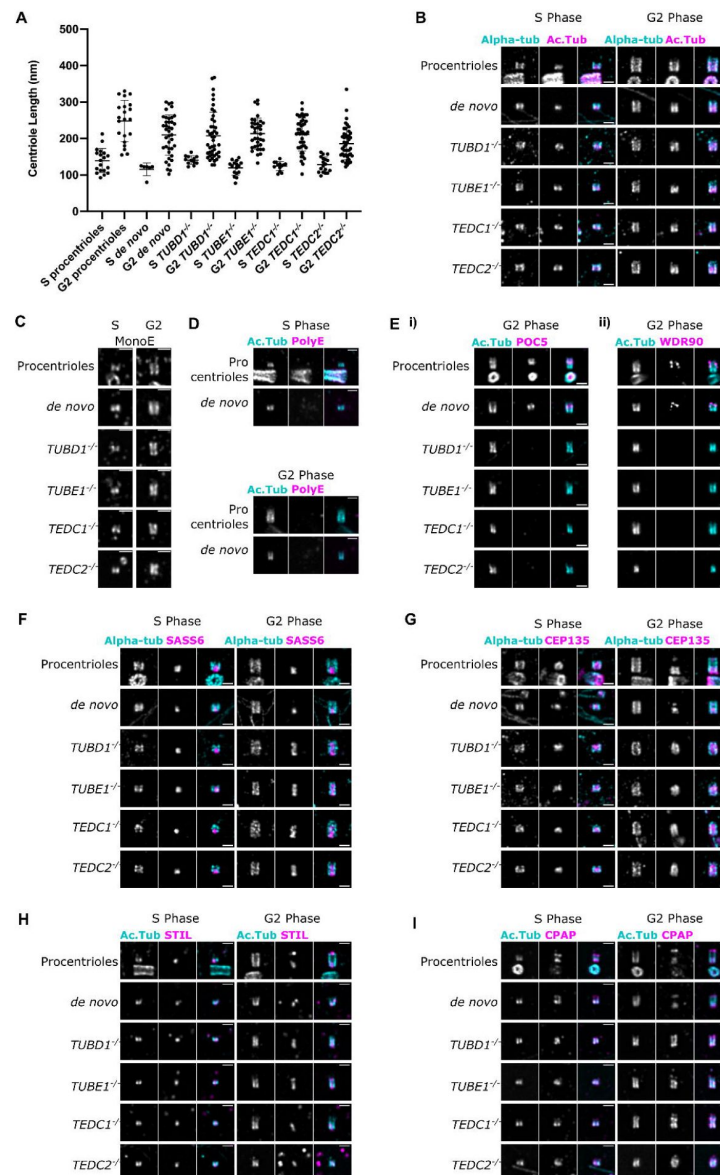


Figure 4.

Mutant centrioles elongate in G2 but fail to recruit central core proteins and have an expanded proximal region

(A) Lengths of expanded centrioles from cells of the indicated cell cycle stages. Lengths were adjusted for the gel expansion factors. Cells were synchronized in S/G2/M and S-phase cells were marked with PCNA. For each genotype, the differences between S and G2 phase centriole lengths are statistically significant (<0.0001 , Welch's t-test). (B) U-ExM images of centrioles stained for alpha-tubulin and acetylated tubulin. (C) U-ExM of centrioles in S or G2 phase stained with monoE (GT335) antibody. (D) U-ExM of control centrioles in S or G2 phase stained with acetylated tubulin and polyE antibodies (E i) U-ExM of centrioles in G2 phase stained with acetylated tubulin (cyan) and POC5 (magenta) antibodies. POC5 is present in the central core of control procentrioles and *de novo* centrioles and absent from mutants. (ii) U-ExM of centrioles in G2 phase stained with acetylated tubulin (cyan) and WDR90 (magenta) antibodies. WDR90 is present in the central core of control procentrioles and *de novo* centrioles, and absent from mutants. (F, G, H, I) U-ExM of centrioles in S and G2 phase stained for alpha tubulin (cyan) or acetylated tubulin (Ac.Tub, cyan) and the following antibodies in magenta: F) SASS6, G) CEP135, H) STIL, I) CPAP. In control centrioles, these proteins are limited to the proximal end. In mutant centrioles, these proteins are present at the proximal end in S phase centrioles and elongate throughout the entire centriole in G2 phase. Images were acquired with a Yokogawa CSU-W1 SoRa with 2.8x relay and deconvolved with 10 iterations using Microvolution. Scale bars: 1 μ m.

Next, we tested whether the centriole proximal end might be properly formed in mutant centrioles. We found that the centriolar cartwheel protein, SASS6, was present within the lumen of control and mutant centrioles in S-phase. In control centrioles in G2-phase, SASS6 was restricted to just the proximal end. Surprisingly, SASS6 was elongated in all G2-phase mutant centrioles (**Fig 4F**, **Fig 4 - Supp 4**). We observed a similar phenotype with multiple other proximal-end proteins: CEP135, STIL, CPAP, and CEP44 (**Fig 4G-I**, **Fig 4 - Supp 1**, **Fig 4 - Supp 3**), indicating that the entire proximal end is elongated in mutant centrioles. The extended localization of proximal end proteins was not due to increased protein expression in mutant cells (**Fig 4 - Supp 2**). We conclude that loss of TEDC1, TEDC2, TUBD1, or TUBE1 results in elongated proximal end domains within mutant centrioles.

Elongation of the proximal end of centrioles may also indicate an overall defect in centriole polarity. To test this hypothesis, we next determined whether these mutant centrioles might properly recruit proteins to their distal ends. We found that CETN2 and CP110, two proteins of the distal centriole, were localized to mutant centrioles and clearly marked one end of the centriole barrel in both S-phase and G2-phase (**Fig 4 - Supp 1B**, **1C**). We conclude that proximal-to-distal centriole polarity was unaffected in mutant centrioles, and proximal end elongation did not affect the recruitment of proteins to the centriole distal end. Together, these results indicate that centrioles lacking compound microtubules are unable to properly regulate the length of the proximal end.

Mutant centrioles elongate further in mitosis before fragmenting

Centrioles lacking triplet microtubules undergo a futile cycle of formation and disassembly, but the mechanisms underlying disassembly are not well-understood. We first tested whether centriole loss in mutant centrioles may be due to loss of CEP295. CEP295 promotes centriole-to-centrosome conversion, a process in which pericentriolar material is recruited to newly-formed procentrioles. Cells lacking CEP295 form centrioles that disintegrate during the cell cycle due to a failure to undergo centriole-to-centrosome conversion (Izquierdo et al., 2014). Using U-ExM, we found that CEP295 was present and normally localized within mutant centrioles in both S- and G2-phases (**Fig 4 - Supp 1F**). We conclude that centriole loss in our mutants is unlikely to be due to loss of CEP295 localization, and therefore that TEDC1, TEDC2, TUBD1 and TUBE1 are likely part of a different pathway required for centriole stability through the cell cycle.

Next, we used U-ExM to visualize centriole loss during mitosis. We stained for the centriole wall (GT335), the centriole proximal end (SASS6) and the centriole distal end (CP110). In control cells, in which centrioles formed *de novo* after centrinone washout, multiple centrioles could be seen throughout mitosis, and SASS6 was lost from centrioles in anaphase-stage cells (**Fig 5A,B**). By contrast, in prometaphase stage *TUBD1*^{-/-} or *TUBE1*^{-/-} cells, we found that centrioles had a unique appearance: they were longer than normal, with an elongated proximal end marked by SASS6, and a CP110-positive cap. These two ends were connected by weak monoE staining (**Fig 5C**, **5E**). This phenotype is identical to our observations of centrioles in a prometaphase *TUBE1*^{-/-} cell by TEM in our previous publication (Wang et al., 2017, **Fig 2B**). After metaphase, centrioles in mutant cells were either completely absent, or had a fragmented appearance (**Fig 5D**, **5F**), with aggregates of staining that did not resemble true centrioles. We conclude that in our mutant cells, centrioles elongate in early mitosis to form an aberrant intermediate structure, followed by fragmentation in late mitosis.

Discussion

Here, we extend our previous study on delta-tubulin (TUBD1), epsilon-tubulin (TUBE1) and the centriolar triplet microtubules. Previously, we showed that loss of either of these proteins from mammalian cultured cell lines results in the same phenotype: loss of the triplet microtubules and

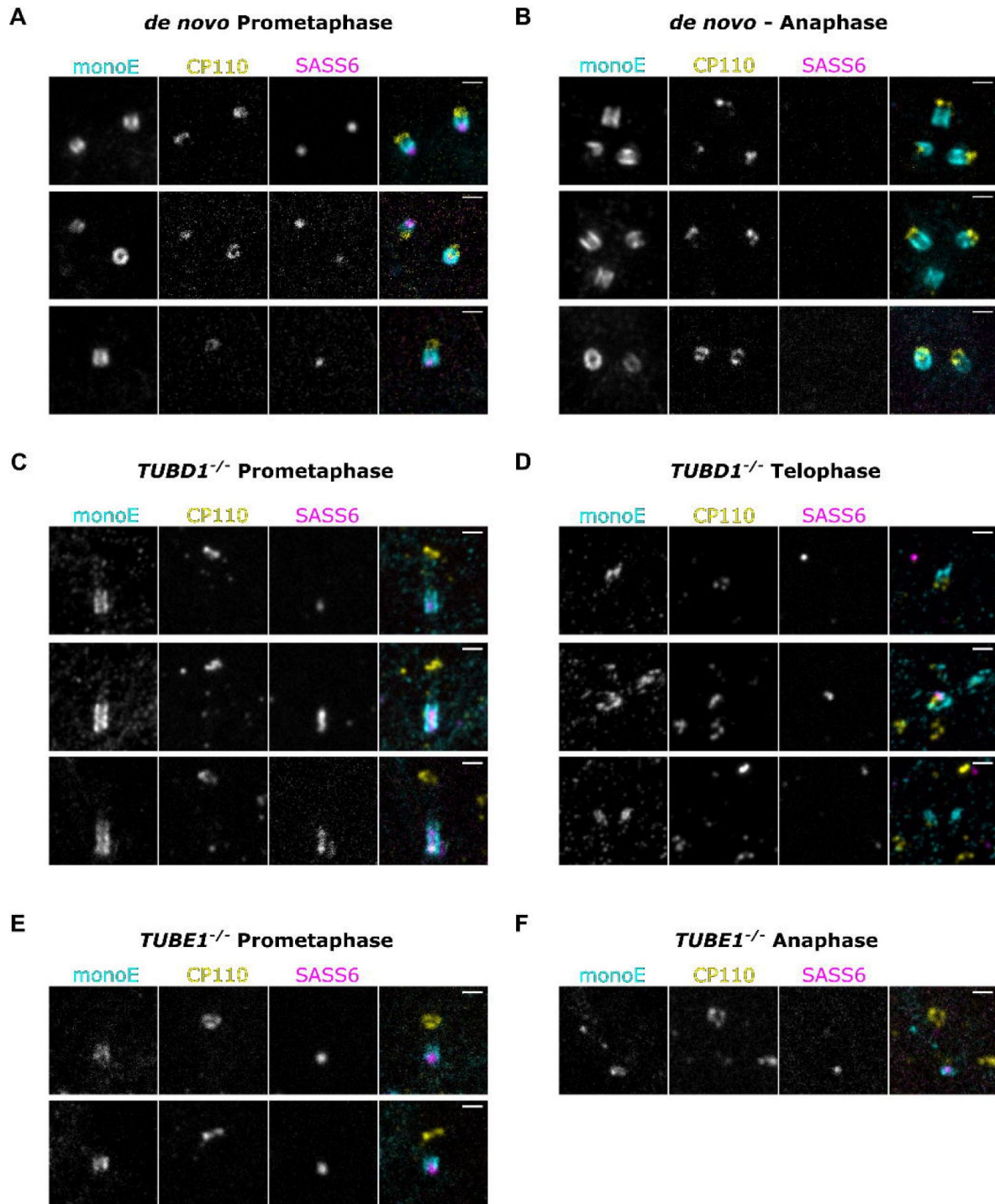


Fig 5.

Mutant centrioles elongate further in mitosis before fragmenting

U-ExM of centrioles stained for monoE (GT335, cyan), CP110 (yellow) and SASS6 (magenta). **(A)** A prometaphase cell with centrioles formed *de novo* **(B)** An anaphase cell with centrioles formed *de novo* **(C)** A prometaphase *TUBD1*^{-/-} cell **(D)** A telophase *TUBD1*^{-/-} cell **(E)** A prometaphase *TUBE1*^{-/-} cell **(F)** An anaphase *TUBE1*^{-/-} cell. Scale bars: 1 μm. Images were acquired with a Yokogawa CSU-W1 SoRA with 2.8x relay.

a futile cycle of centriole formation and disintegration (Wang et al., 2017 [DOI](#)). Here, we add two new proteins to this pathway: TEDC1 and TEDC2, which were originally identified by their association with TUBD1 and TUBE1 (Breslow et al., 2018 [DOI](#); Huttlin et al., 2017 [DOI](#); Huttlin et al., 2021 [DOI](#)). Loss of TEDC1 or TEDC2 phenocopies the loss of TUBD1 or TUBE1: aberrant centrioles are formed that lack triplet microtubules and disintegrate during passage through mitosis. TEDC1 and TEDC2 localize to centrioles, indicating that they have a direct role in forming or maintaining centriole structure, and their localization depends on each of the other three proteins within the complex. All four proteins physically interact with each other. Using our mutant cell lines, we interrogated whether any of these proteins can form subcomplexes within cells. We found that TEDC1 and TEDC2 can interact with each other independently of the tubulins, supporting a predicted AlphaFold-Multimer model. Together, these results indicate that these four proteins act together in a complex at centrosomes to form or stabilize the compound microtubules.

While the molecular mechanisms underlying the function of this complex are unknown, an attractive model is that the tetrameric complex forms a structural component of centrioles. Our AlphaFold models indicate that such a structure would be approximately 13 nm in length and 6 nm in width. Within procentrioles and the proximal region of the parental centriole, it is possible that these four proteins help form the A-C linker, the pinhead, or the triplet base. Recently, components of the A-C linker have been identified (Bournonville et al., 2024 [DOI](#); Laporte et al., 2024 [DOI](#)), and three of the proteins in our complex (TEDC2, TUBD1, and TUBE1) had shared co-dependencies with A-C linker components using DepMap (Bournonville et al., 2024 [DOI](#)). The A-C linker is lost from our mutant centrioles, but it is not clear whether this is because these proteins have a direct role in forming A-C linkers or whether this reflects an indirect role of the triplet microtubules in stabilizing A-C linkers. We note that it is also possible that only some proteins of the complex, such as delta-tubulin and epsilon-tubulin, form structural components of centrioles, or that the complex may interact transiently with centrioles. Future experiments will reveal the mechanisms by which these proteins act.

Using ultrastructure expansion microscopy, we find that mutant centrioles with singlet microtubules exhibit additional major architectural defects, including absence of the inner scaffold and elongation of the proximal end. We propose that the absence of the inner scaffold arises from the loss of the B- and C-tubules within centrioles, which may serve to anchor WDR90 and/or other proteins of the inner scaffold. WDR90 has been proposed to localize to the inner junction between the A- and B-tubules and is required for recruiting other inner scaffold components (Le Guennec et al., 2020 [DOI](#); Steib et al., 2020 [DOI](#)). We find that mutant centrioles with singlet microtubules fail to localize WDR90, and thus speculate that the B-tubule is required to recruit or stabilize WDR90 at the inner junction. In addition, by cryo-electron tomography, the inner scaffold makes connections to all three (A-, B-, and C-) tubules. Though the identities of all the proteins that form these connections have not been determined, it is possible that mutant centrioles with only A-tubules also fail to provide anchoring sites for the other proteins within the inner scaffold. Together, these results demonstrate that the compound microtubules of centrioles are required for proper formation of the inner helical scaffold of the central core.

Mutant centrioles with singlet microtubules have an elongated proximal end that extends the entire length of the centriole, as marked by multiple proximal end markers (SASS6, CEP135, STIL, CPAP, CEP44). These results are also supported by our previous observations that by TEM, the lumen of *TUBD1*^{-/-} and *TUBE1*^{-/-} mutant centrioles are filled with electron-dense material (Wang et al., 2017 [DOI](#)). Little is known about the molecular mechanisms that regulate proximal end length, though centrioles from the symbiotic flagellate *Trichonympha* bear an elongated proximal region with extended cartwheel, and the doublet and singlet-bearing centrioles from *Drosophila* and *C. elegans* have cartwheels that extend the entire length of the centriole (González et al., 1998 [DOI](#); Guichard and Gönczy, 2016 [DOI](#); Guichard et al., 2012 [DOI](#); Pelletier et al., 2006 [DOI](#); Woglar et al., 2022 [DOI](#)). It is possible that the triplet microtubules, the inner scaffold, and/or the TUBD1/TUBE1/TEDC1/TEDC2 protein complex might act to limit the length of the proximal end.

Recently, loss of the inner scaffold protein POC1A has been shown to result in centrioles with extended regions of some proximal proteins, including CEP44, CEP135, and CEP295, indicating that the inner scaffold regulates the extent of these proteins (Sala et al., 2024 [DOI](#)). Interestingly, unlike our mutant centrioles which have singlet microtubules, *POC1A*^{-/-} mutant centrioles can form triplet microtubules and do not have extended SASS6 staining (Sala et al., 2024 [DOI](#)). This suggests that the height of the cartwheel may be regulated by the triplet microtubules. The cartwheel and centriolar microtubules have been proposed to assemble interdependently to impart ninefold symmetry upon the centriole (Hilbert et al., 2016 [DOI](#)), and it is possible that interdependent assembly also regulates the height of the cartwheel.

Many aspects of centriole architecture, including formation of the distal tip, centriole length regulation prior to mitosis, acquisition of post-translational modifications, establishment of proximal-distal polarity, and recruitment of proteins required for centriole-to-centrosome conversion, are unaffected in mutant centrioles. These results indicate that the proteins that regulate these processes can act upon the A-tubule independently of the B- and C-tubules.

Here, we also extend our previous observations of centriole loss in mutant centrioles. In most cell types, centrioles are inherited by daughter cells during each mitosis. Centriole loss is not unique to centrioles lacking compound microtubules: mammalian cells engineered to lack CEP295 also form centrioles that are lost through the cell cycle, due to an inability to undergo centriole to centrosome conversion (Izquierdo et al., 2014 [DOI](#)). Similarly, in *Drosophila* oocytes, down-regulation of Polo kinase and pericentriolar material triggers centriole elimination (Pimenta-Marques et al., 2016 [DOI](#)). We find that CEP295 is properly localized in mutant centrioles with singlet microtubules, indicating that centriole loss in this context may be independent of centriole to centrosome conversion and pericentriolar material recruitment. Using expansion microscopy, we find that centriole loss is correlated with loss of the SASS6 cartwheel in mitosis. In this regard, mutant centrioles with singlet microtubules resemble centriole loss within *C. elegans* oocytes, in which an analogous structure to the cartwheel named the central tube is lost prior to centriole widening and subsequent loss of the centriolar microtubules (Pierron et al., 2023 [DOI](#)). In addition, centriole loss in our mutant cells occurs through a stereotyped progression of architectural changes in mitosis, starting with centriole over elongation in prometaphase and culminating with centriole fragmentation and loss. Prolonged mitotic arrest has been reported to result in centriole over elongation through Plk1 activity (Kong et al., 2020 [DOI](#)), and it is possible that a lengthened mitosis, as observed in these mutant cells and cells lacking centrioles (Farrell et al., 2024 [DOI](#); Wang et al., 2017 [DOI](#)), may also result in over elongation of mutant centrioles with just A-tubules. In addition, we note that CPAP has an expanded domain in mutant centrioles compared to controls (Fig 4 [DOI](#), (Vásquez-Limeta et al., 2022 [DOI](#)). CPAP is involved in slow processive microtubule growth (Sharma et al., 2016 [DOI](#)) and its loss results in centriole fragmentation (Vásquez-Limeta et al., 2022 [DOI](#)), and it is possible that CPAP mislocalization may also contribute to over elongation of these mutant centrioles. Future work will determine the molecular mechanisms by which mutant centrioles lacking triplet microtubules are disassembled through the cell cycle.

Finally, we note that mutant human centrioles lacking compound microtubules bear similarities to the centrioles of *Drosophila* and *C. elegans* embryos, which have evolved to lack triplet microtubules and have cartwheels extending the entire length of the centriole (González et al., 1998 [DOI](#); Pelletier et al., 2006 [DOI](#); Woglar et al., 2022 [DOI](#)). Embryonic centrioles in both species are shorter than that of other organisms, and helical inner scaffolds have not been reported. In both species, these diminished centrioles participate in mitosis, can duplicate their centrioles, and serve as basal bodies for sensory cilia. We speculate that centrioles with triplet microtubules and the proteins they anchor, including the inner scaffold, may be required for centriole function in organisms with motile cilia, perhaps to help stabilize the basal body against ciliary movement. Such activity has been described for *Tetrahymena* basal bodies, and mutating an inner scaffold protein, Poc1, results in abnormal bending within basal bodies (Junker et al., 2022 [DOI](#)). Further supporting this hypothesis, *Drosophila* spermatocytes, one of the few cells within this species with

motile cilia, have basal bodies with triplet microtubules (González et al., 1998 [↗](#)). We note that these spermatocytes likely form triplet microtubules in an alternative manner, as *Drosophila* lacks delta-tubulin or epsilon-tubulin.

In conclusion, this work, along with our previously published study, identifies proteins required for the formation or maintenance of the centriolar triplet microtubules and maps the requirements of these proteins and the triplets in centriole architecture. Together, these results pave the way for deeper molecular understanding of the mechanisms by which the triplet microtubules are formed and maintained reproducibly within cells to form robust centrioles and cilia.

Figure legends

Materials and Methods

Cell lines and cell culture

hTERT RPE-1 *TP53*^{-/-} cells were a gift from Meng-Fu Bryan Tsou (Memorial Sloan Kettering Cancer Center) and were cultured in DMEM/F-12 (Corning) supplemented with 10% Cosmic Calf Serum (CCS; HyClone). HEK293T cells for lentivirus production (see below) were obtained from the ATCC and cultured in DMEM (Corning) supplemented with 10% CCS. hTERT RPE-1 and HEK293T/17 cells were authenticated using STR profiling using CODIS loci. All other cell lines used were derived from hTERT RPE-1 *TP53*^{-/-} cells. Stable *TP53*^{-/-}; *TEDC1*^{-/-} and *TP53*^{-/-}; *TEDC2*^{-/-} knockout cell lines were made in the hTERT RPE-1 *TP53*^{-/-} cells by CRISPR/Cas9 (see below). For rescue experiments, clonal knockout cell lines were rescued using lentiviral transduction (see below). All cells were cultured at 37°C under 5% CO₂, and are mycoplasma-free (Uphoff and Drexler, 2011 [↗](#)).

Generation of *TEDC1*^{-/-} and *TEDC2*^{-/-} cells and rescue cell lines

TEDC1^{-/-} and *TEDC2*^{-/-} cells were generated by CRISPR/Cas9 mediated gene editing using a recombinantly produced, purified Cas9 protein (Cas9-NLS, QB3 Macrolab, Berkeley) and chemically synthesized two-component gRNA (crRNA:tracrRNA, Alt-R CRISPR-Cas9 system, IDT). For increased efficiency, two gRNAs, both targeting the 5' end of each gene, were used at the same time. Target sequences were: 5'-CGCCAAGTTCGACCGTCCGG-3' and 5'-CGTCCAATCACCGCACGGGC-3' for *TEDC1*, and 5'-CGCACAGCGACAATTGCAAT-3' and 5'-CACCGGCGGAGAGCCCCGC-3' for *TEDC2*.

Lyophilized RNA oligos were reconstituted according to the instructions provided by the manufacturer (IDT). Briefly, oligos were reconstituted in the duplex buffer at a concentration of 200 μM. To anneal crRNA with tracrRNA, 3 μl of each (600 pmol) were mixed, heated to 95°C, and transferred to room temperature to gradually cool. Pre-complexed crRNA and tracrRNA (550 pmol) were mixed with purified Cas9 (360 pmol), diluted with PBS to a total volume of 25 μl and incubated for 15 min at room temperature to form ribonucloprotein complexes (RNPs).

RPE1 *TP53*^{-/-} cells stably expressing GFP-centrin (Wang et al., 2017 [↗](#)) were electroporated in a home-made electroporation buffer (Zhang et al., 2014 [↗](#)) using Amaxa Nucleofector II (Lonza). Cells were electroporated with an equal mix of two RNPs: 50 μl of RNPs mixture was added to 2 x 10⁶ cells in 200 μl electroporation buffer. To facilitate the identification of electroporated cells, an mRuby2 expressing plasmid (pcDNA3-mRuby2, plasmid pTS3994) was electroporated together with RNPs.

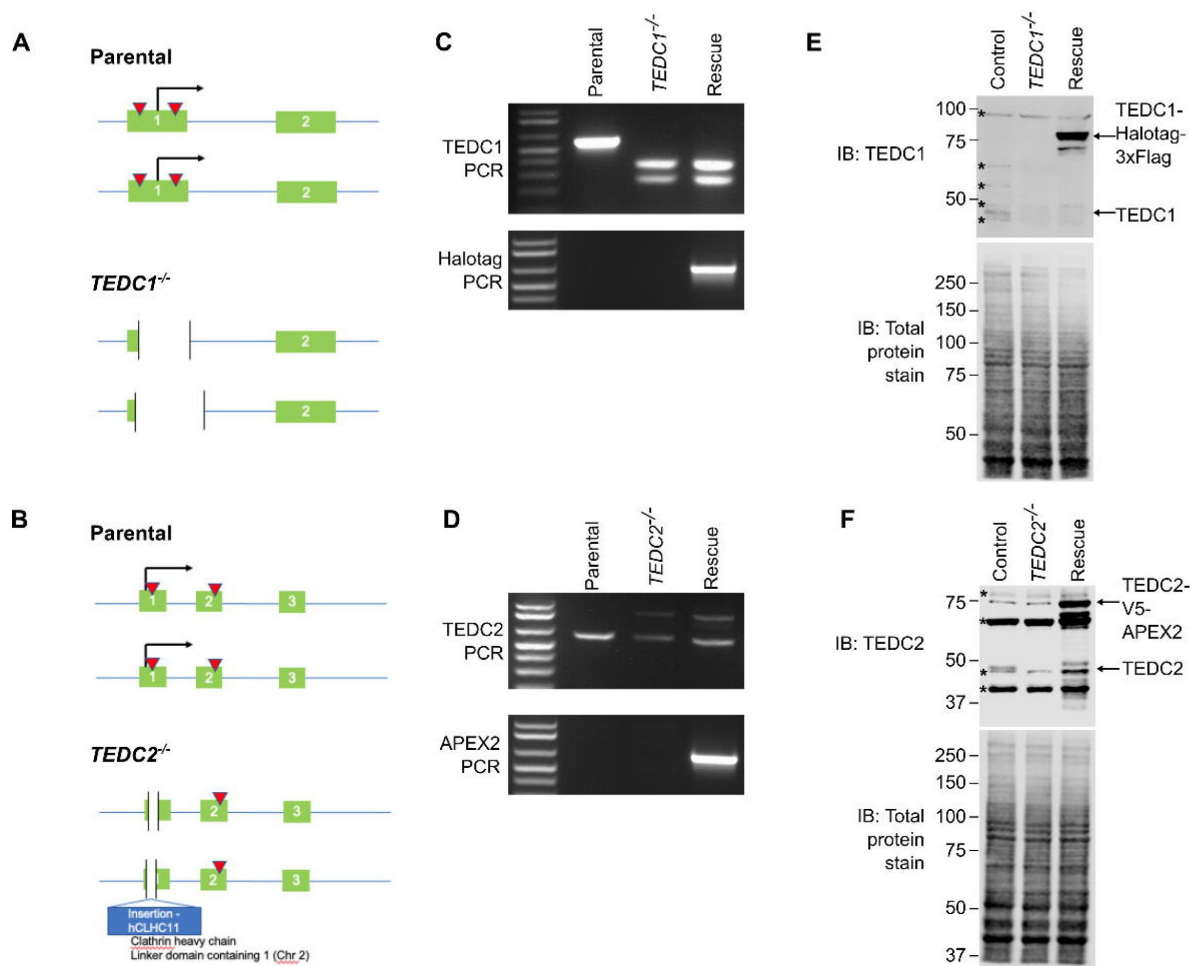


Figure 1 - Supplementary Figure 1.

Genotyping of *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant cell lines

(A) Gene structure of the *TEDC1* locus in parental *TP53*^{-/-} cells and the *TEDC1*^{-/-} mutant. Green boxes: exons; blue lines: introns; red triangles: sgRNA binding sites; black arrow: translation start site. The *TEDC1*^{-/-} mutant (clone 2F4) is a compound heterozygote bearing a deletion of 227 bp on one allele and a deletion of 329 bp on the other allele. In both alleles, the ATG start site is deleted and the next ATG is not in-frame. **(B)** Gene structure of the *TEDC2* locus in parental *TP53*^{-/-} cells and the *TEDC2*^{-/-} mutant. Green boxes: exons; blue lines: introns; red triangles: sgRNA binding sites; black arrow: translation start site. The *TEDC2*^{-/-} mutant (clone F5) is a compound heterozygote bearing a deletion of 19 bp on one allele flanking the ATG start site. On the other allele, there is an insertion of 306 bp corresponding to a fusion between *TEDC2* and the hCLHC1 gene. In both alleles, the ATG start site is deleted, the next ATG is not in-frame, and no additional ATG start sites are found. **(C)** Genotyping PCR of the *Tedc1* locus in parental *TP53*^{-/-} cells, the *TEDC1*^{-/-} mutant, and *TEDC1* Rescued (RPE1 *TP53*^{-/-}; *TEDC1*^{-/-}; *TEDC1*-Halotag-3xflag) cells. Top: PCR for *TEDC1*. Bottom: PCR for Halotag. **(D)** Genotyping PCR of the *TEDC2* locus in parental *TP53*^{-/-} cells, the *TEDC2*^{-/-} mutant, and *TEDC2* Rescued (RPE1 *TP53*^{-/-}; *TEDC2*^{-/-}; *TEDC2*-V5-APEX2) cells. Top: PCR for *TEDC2*. Bottom: PCR for APEX2. **(E)** Western blot of *TEDC1* protein levels in parental *TP53*^{-/-} cells, the *TEDC1*^{-/-} mutant, and *TEDC1* Rescued (RPE1 *TP53*^{-/-}; *TEDC1*^{-/-}; *TEDC1*-Halotag-3xflag) cells. Total protein stain is used as a loading control. *TEDC1*-Halotag-3xFlag is overexpressed 73-fold above endogenous levels (average of 3 independent experiments). Asterisks mark non-specific bands. **(F)** Western blot of *TEDC2* protein levels in parental *TP53*^{-/-} cells, the *TEDC2*^{-/-} mutant, and *TEDC2* Rescued (RPE1 *TP53*^{-/-}; *TEDC2*^{-/-}; *TEDC2*-V5-APEX2) cells. Total protein stain is used as a loading control. *TEDC2*-V5-APEX2 is overexpressed 26-fold above endogenous levels (average of 3 independent experiments). Asterisks mark non-specific bands.

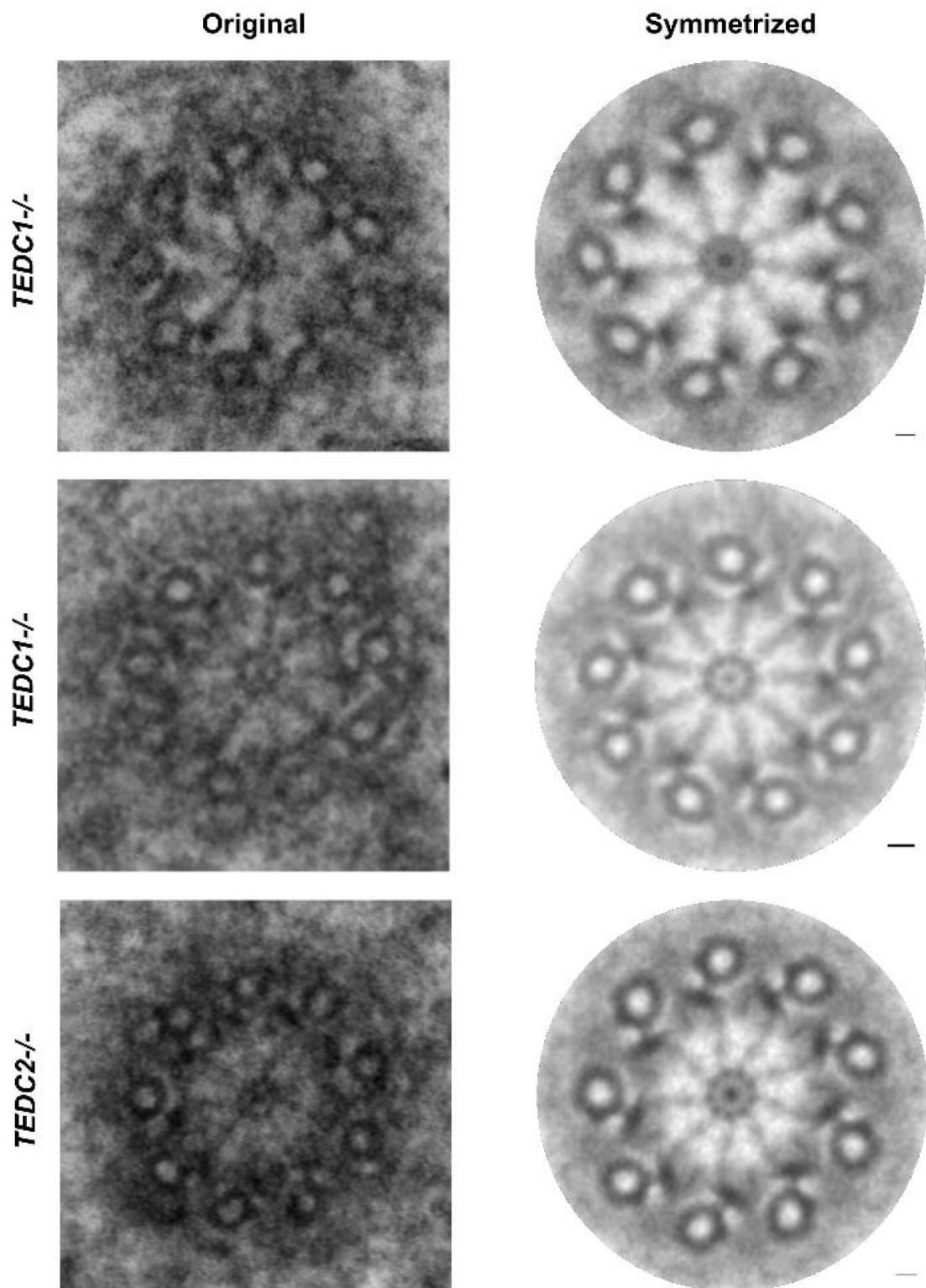


Figure 1 - Supplementary Figure 2.

Symmetrization of *TEDC1*^{-/-} and *TEDC2*^{-/-} mutant centrioles

Original (left) and symmetrized (right) images of TEM images of *TEDC1*^{-/-} and *TEDC2*^{-/-} centrioles. The first image is the same as that in [Fig 1E](#), the last image is the same as that in [Fig 1F](#). The middle image is an additional centriole from the *TEDC1*^{-/-} mutant cells.

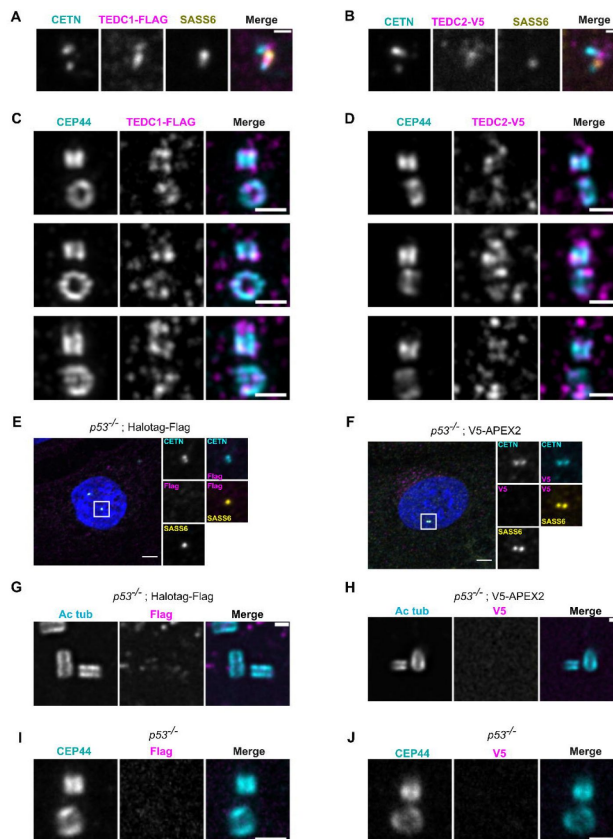


Figure 2 - Supplementary Figure 1.

Extended localization analyses of TEDC1 and TEDC2

(A) Immunofluorescence staining of a TEDC1 rescue cell in G2 phase expressing TEDC1-Halotag-3xFlag, super-resolution image using SoRA disk and 2.8x relay. Maximum projection. Cyan: Centrin (CETN); Magenta: TEDC1-Halotag-3xFlag (localized with anti Flag antibody); Yellow: SASS6. Scale bar: 0.5 μ m **(B)** Immunofluorescence staining of a TEDC2 rescue cell in G2 phase expressing TEDC2-V5-APEX2, super-resolution image using SoRA disk and 2.8x relay. Maximum projection. Cyan: Centrin (CETN); Magenta: TEDC2-V5-APEX2 (localized with anti V5 antibody); Yellow: SASS6. Scale bar: 0.5 μ m **(C)** Expansion microscopy image of TEDC1 rescue cells expressing TEDC1-Halotag-3xFlag. Expansion gel was made as described in (Kong et al., 2024). The procentriole is oriented vertically. Cyan: CEP44; Magenta: TEDC1-Halotag-3xFlag (localized with anti-Flag antibody). Deconvolved using Microvolution; maximum projection. Scale bar: 1 μ m. **(D)** Expansion microscopy image of TEDC2 rescue cells expressing TEDC2-V5-APEX2. Expansion gel was made as described in (Kong et al., 2024). The procentriole is oriented vertically. Cyan: CEP44; Magenta: TEDC2-V5-APEX2 (localized with anti-V5 antibody). Deconvolved using Microvolution; maximum projection of confocal stacks. Scale bar: 1 μ m. **(E)** Immunofluorescence staining of *TP53*^{-/-} cells expressing Halotag-Flag - negative control for Fig 2A. Images are maximum projections of confocal stacks and were acquired with the same exposure settings as in Fig 2A. Blue: DAPI; Cyan: Centrin; Magenta: Flag; Yellow: SASS6. Scale bar: 5 μ m. **(F)** Immunofluorescence staining of *TP53*^{-/-} cells expressing V5-APEX2 - negative control for Fig 2B. Images are maximum projections of confocal stacks and were acquired with the same exposure settings as in Fig 2B. Blue: DAPI; Cyan: Centrin (localized with anti-GFP antibody recognizing GFP-centrin); Magenta: V5; Yellow: SASS6. Scale bar: 5 μ m. **(G)** U-ExM of *TP53*^{-/-} cells expressing Halotag-Flag - negative control for Fig 2C. Cyan: Acetylated tubulin; Magenta: Flag. Confocal image stacks were deconvolved using Microvolution; single plane images shown. Images were acquired using the same parameters as Fig 2C. Scale bar: 1 μ m. **(H)** U-ExM of *TP53*^{-/-} cells expressing V5-APEX2 - negative control for Fig 2D. Cyan: Acetylated tubulin; Magenta: V5. Confocal image stacks were deconvolved using Microvolution; single plane images shown. Images were acquired using the same parameters as Fig 2D. Scale bar: 1 μ m. **(I)** Expansion microscopy image of *TP53*^{-/-} cells stained with Flag antibody, negative control for Fig 2 - Supp 1C. Cyan: CEP44; Magenta: Flag. Confocal image stacks were deconvolved using Microvolution; image is a maximum projection of confocal stack. Scale bar: 1 μ m. **(J)** Expansion microscopy image of *TP53*^{-/-} cells stained with V5 antibody, negative control for Fig 2 - Supp 1D. Cyan: CEP44; Magenta: V5. Confocal image stacks were deconvolved using Microvolution; image is a maximum projection of confocal stack. Scale bar: 1 μ m.

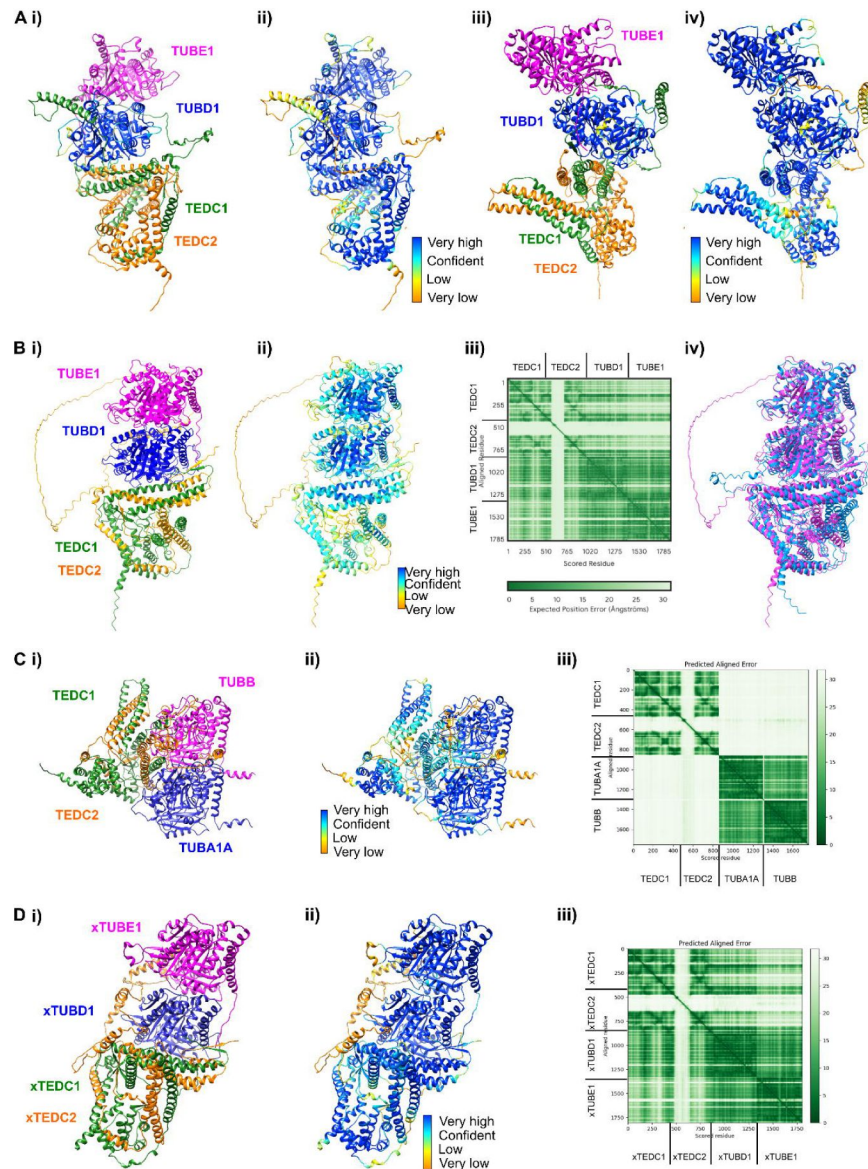


Fig 3 - Supplementary Figure 1.

AlphaFold-Multimer and AlphaFold3 predictions (Ai)

Rotated view of the AlphaFold-Multimer prediction from **Fig 3E** (120 degrees around the y- axis) **(Aii)** Rotated view colored according to pLDDT. Very high: pLDDT > 90. High: 90 > pLDDT > 70. Low: 70 > pLDDT > 50. Very low: pLDDT < 50 **(Aiii)** Rotated view of the AlphaFold-Multimer prediction from **Fig 3E** (240 degrees around the y-axis) **(Aiv)** Rotated view colored according to pLDDT. Very high: pLDDT > 90. High: 90 > pLDDT > 70. Low: 70 > pLDDT > 50. Very low: pLDDT < 50 **(Bi)** AlphaFold3 prediction of the complex **(Bii)** AlphaFold3 prediction colored according to pLDDT. Very high: pLDDT > 90. High: 90 > pLDDT > 70. Low: 70 > pLDDT > 50. Very low: pLDDT < 50 **(Biii)** Predicted align error of the AlphaFold3 prediction. Expected position error (Angstroms) is graphed). **(Biv)** Structural alignment between the AlphaFold3 prediction (magenta) and the AlphaFold-Multimer prediction (cyan). Using ChimeraX v1.7.1 Matchmaker, the RMSD between 450 pruned atom pairs is 0.538 angstroms (across all 475 pairs: 0.979). **(Ci)** AlphaFold-Multimer prediction of TEDC1, TEDC2, TUBA1A, TUBB **(Cii)** AlphaFold-Multimer prediction from Ci) colored according to pLDDT. Very high: pLDDT > 90. High: 90 > pLDDT > 70. Low: 70 > pLDDT > 50. Very low: pLDDT < 50 **(Ciii)** Predicted align error of the AlphaFold-Multimer prediction from Ci). Expected position error (Angstroms) is graphed. **(Di)** AlphaFold Multimer prediction of *Xenopus* TEDC1, TEDC2, TUBD1, TUBE1 **(Dii)** AlphaFold-Multimer prediction from Di) colored according to pLDDT. Very high: pLDDT > 90. High: 90 > pLDDT > 70. Low: 70 > pLDDT > 50. Very low: pLDDT < 50 **(Diii)** Predicted align error of the AlphaFold-Multimer prediction from Di). Expected position error (Angstroms) is graphed.

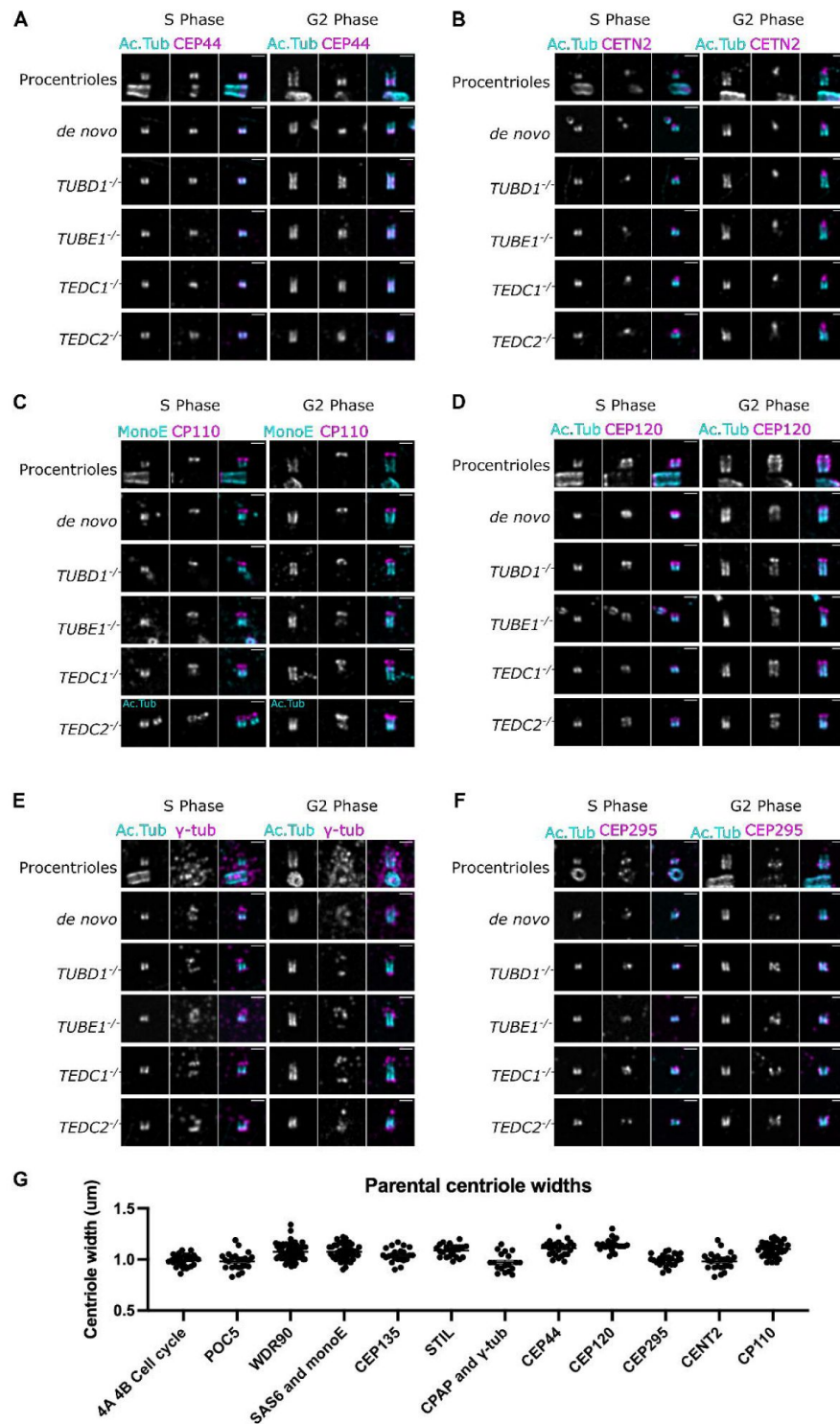


Figure 4 - Supplementary figure 1

Extended analyses of mutant centriole architecture and U-ExM gel expansion factor

(**A, B, C, D, E, F**) U-ExM of centrioles in S and G2 phase stained for acetylated tubulin (cyan) and the following proteins in magenta: **A**) CEP44, **B**) CETN2, **C**) CP110, **D**) CEP120, **E**) gamma-tubulin, **F**) CEP295. Scale bars = 1 μm. Images were acquired with a Yokogawa CSU-W1 SoRA with 2.8x relay and deconvolved with 10 iterations using Microvolution. (**G**) Measurements of the widths of parental centrioles from each experiment as a readout of expansion factor, including the cell cycle analyses in **Fig 4A** and **Fig 4B**. Centriole widths were a mean of 1.0 μm, corresponding to a four-fold expansion factor.

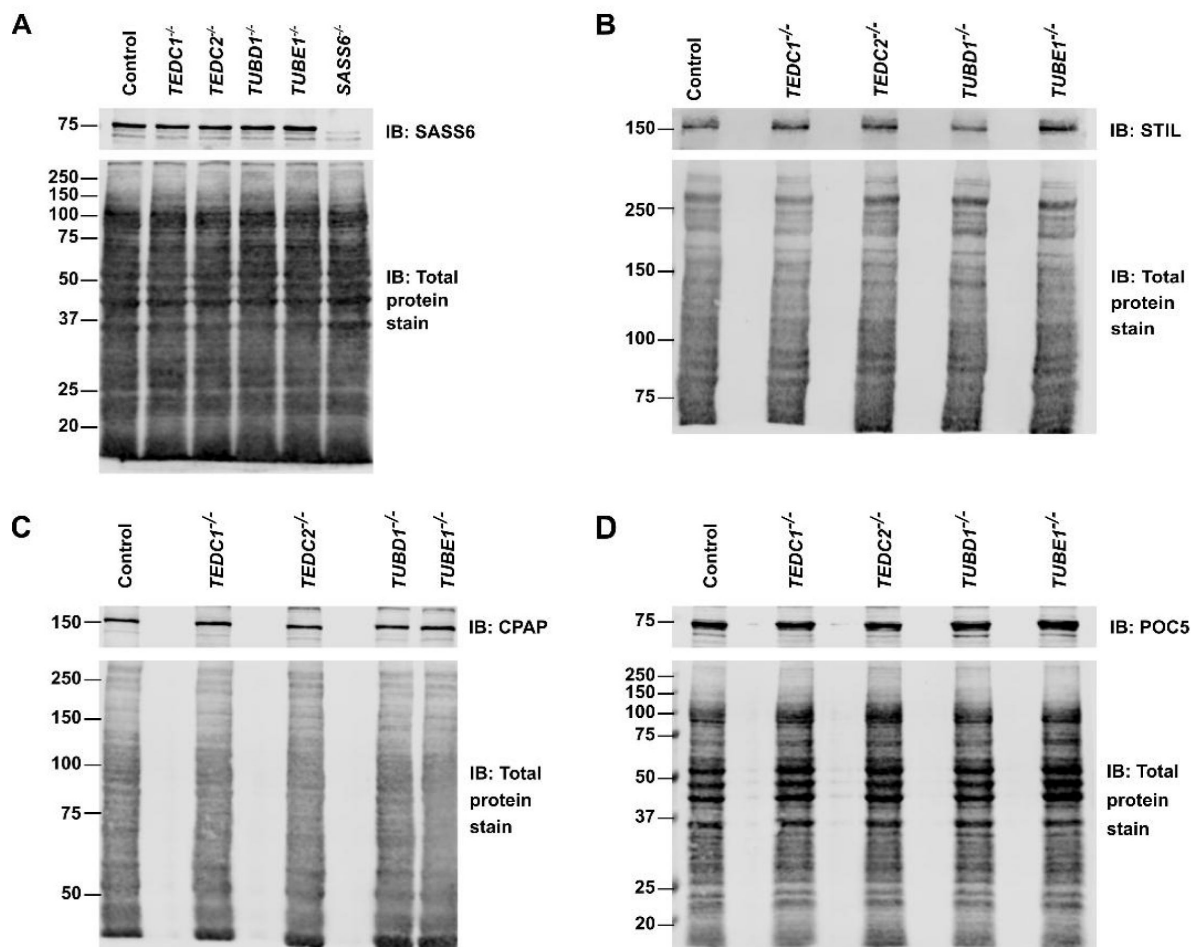


Figure 4 - Supplementary figure 2

Total protein levels of centrosomal proteins are unchanged in mutant cells

(A) Western blot of control (RPE1 *TP53*^{-/-}), *TEDC1*^{-/-}, *TEDC2*^{-/-}, *TUBD1*^{-/-}, *TUBE1*^{-/-}, *SASS6*^{-/-} cell lysates, immunoblotted for SASS6. Total protein stain (Revert) serves as a loading control. **(B)** Western blot of control (RPE1 *TP53*^{-/-}), *TEDC1*^{-/-}, *TEDC2*^{-/-}, *TUBD1*^{-/-}, *TUBE1*^{-/-} cell lysates, immunoblotted for STIL. Total protein stain (Revert) serves as a loading control. **(C)** Western blot of control (RPE1 *TP53*^{-/-}), *TEDC1*^{-/-}, *TEDC2*^{-/-}, *TUBD1*^{-/-}, *TUBE1*^{-/-} cell lysates, immunoblotted for CPAP. Total protein stain (Revert) serves as a loading control. **(D)** Western blot of control (RPE1 *TP53*^{-/-}), *TEDC1*^{-/-}, *TEDC2*^{-/-}, *TUBD1*^{-/-}, *TUBE1*^{-/-} cell lysates, immunoblotted for POC5. Total protein stain (Revert) serves as a loading control.

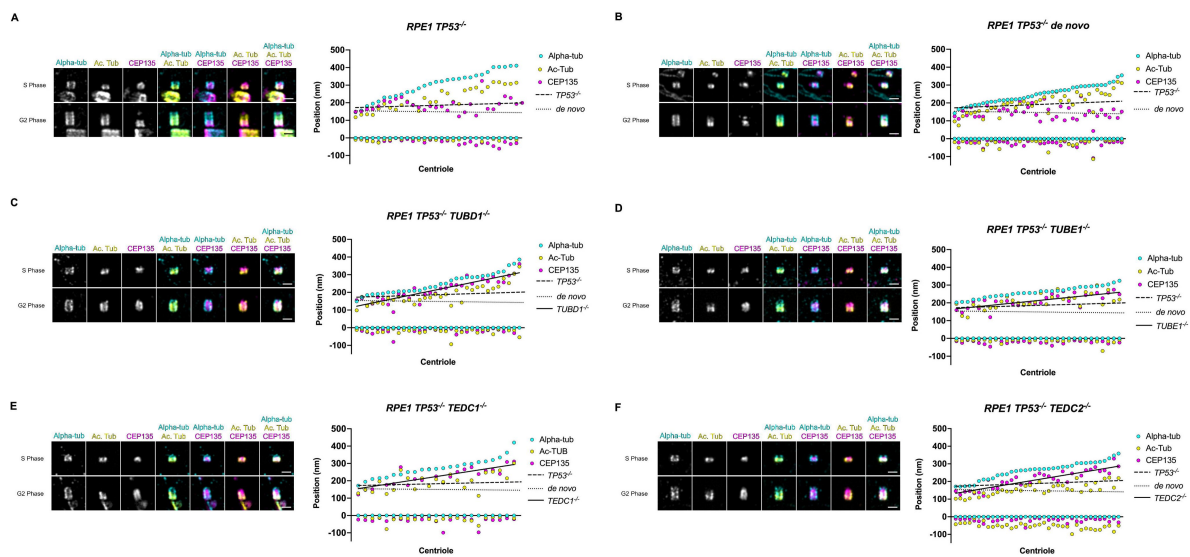


Figure 4 - Supplementary figure 3

Quantification of CEP135 centriolar localization through S and G2 phase

Mutant centrioles have over-elongated CEP135. A) control procenrioles, n=29 centrioles; B) *de novo* centrioles, n=42 centrioles; C) *TUBD1*^{-/-}, n=32 centrioles; D) *TUBB1*^{-/-}, n=30 centrioles; E) *TEDC1*^{-/-}, n=23 centrioles; F) *TEDC2*^{-/-}, n=36 centrioles. For each panel, representative U-ExM images of centrioles in S and G2 phase are shown. These are the same centrioles as in **Figure 4G** and were stained for alpha-tubulin (cyan), acetylated tubulin (yellow), and CEP135 (magenta). Scale bars = 1 μ m. Graphs: Each column represents a centriole, for which the proximal and distal positions of CEP135 (magenta), acetylated tubulin (yellow) and alpha-tubulin (cyan) are displayed. Centrioles were arranged from shortest to longest. Numbers were adjusted for expansion factor. A line of best fit was added for CEP135 position: control procenrioles (dashed), *de novo* centrioles (dotted), and mutants (solid).

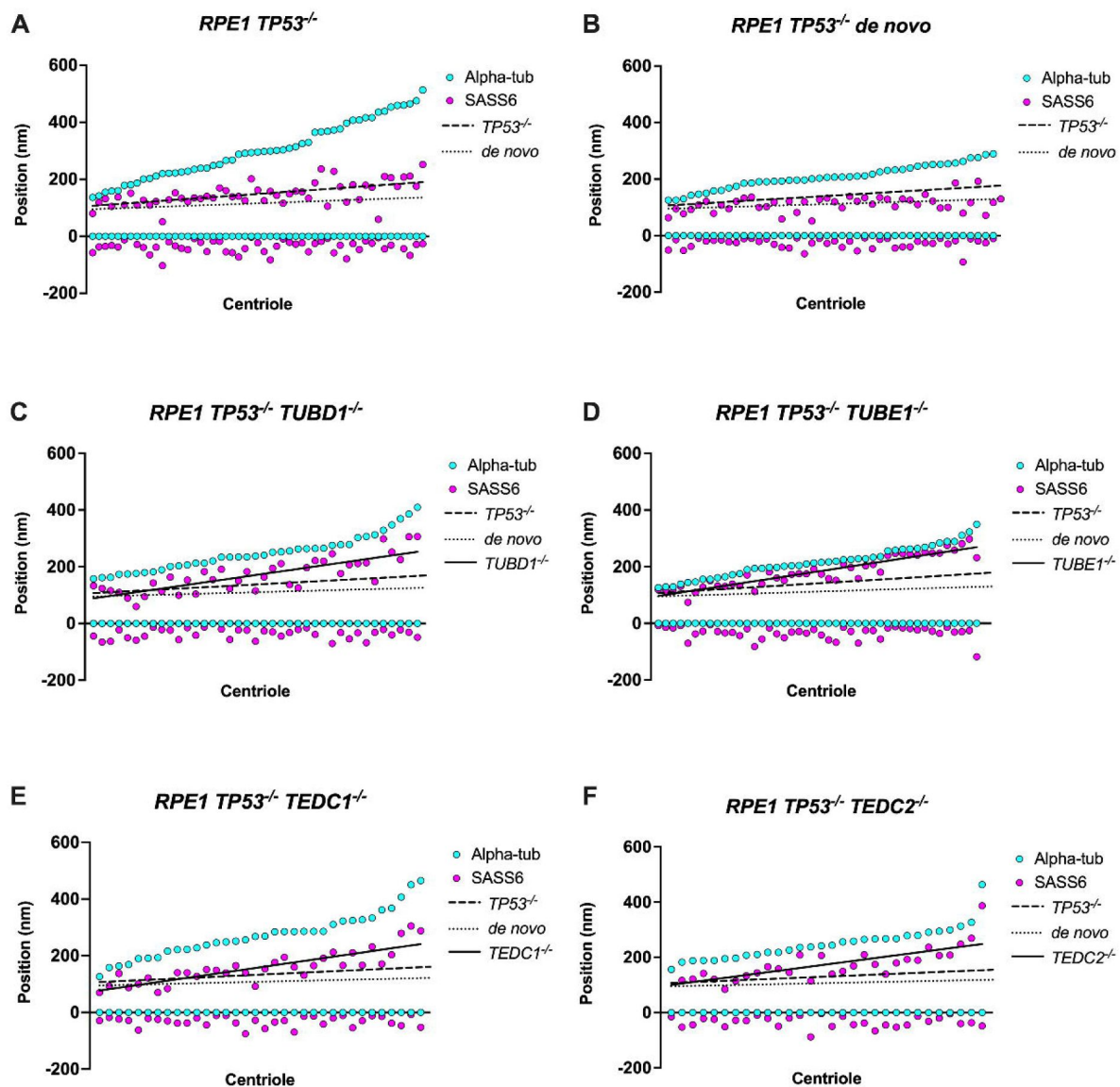


Figure 4 - Supplementary figure 4

Quantification of SASS6 centriolar localization through S and G2 phase

Mutant centrioles have over-elongated SASS6. A) control procenrioles, n=53 centrioles; B) *de novo* centrioles, n=44 centrioles; C) *TUBD1^{-/-}*, n=39 centrioles; D) *TUBE1^{-/-}*, n=44 centrioles; E) *TEDC1^{-/-}*, n=34 centrioles; F) *TEDC2^{-/-}*, n=30 centrioles. Each column represents a centriole, for which the proximal and distal positions of SASS6 (magenta) and alpha-tubulin (cyan) are displayed. Centrioles were arranged from shortest to longest. A Numbers were adjusted for expansion factor. line of best fit was added for SASS6 position: control procenrioles (dashed), *de novo* centrioles (dotted), and mutants (solid).

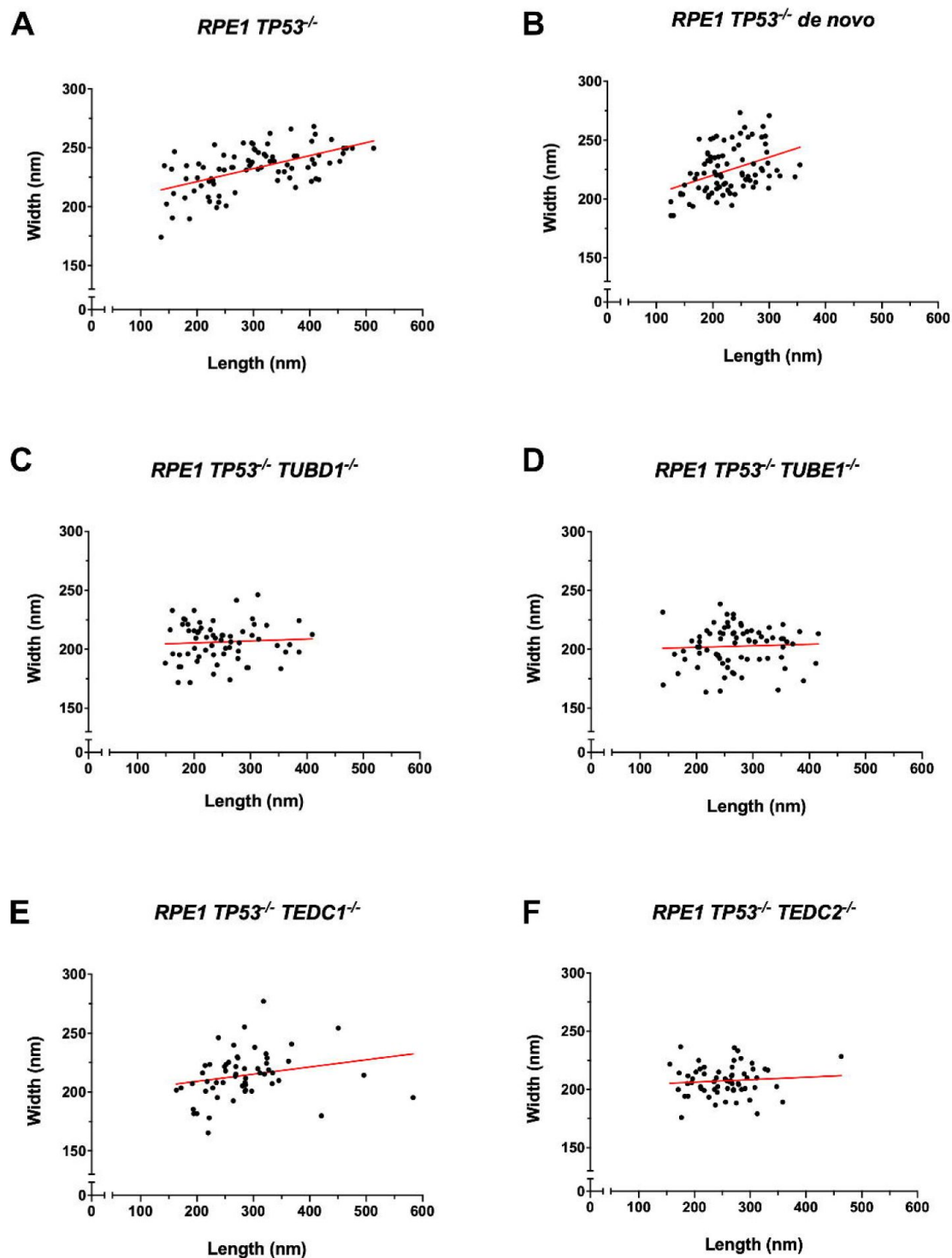


Figure 4 - Supplementary figure 5

Quantification of centriole widths and lengths

Mutant centrioles have smaller widths compared to controls. A) control procentrioles, n=82 centrioles; B) *de novo* centrioles, n=86 centrioles; C) *TUBD1^{-/-}*, n=64 centrioles; D) *TUBE1^{-/-}*, n=74 centrioles; E) *TEDC1^{-/-}*, n=56 centrioles; F) *TEDC2^{-/-}*, n=62 centrioles. Centriole widths and lengths measured by alpha-tubulin antibody are graphed, adjusted for expansion factor. A line of best fit was added (red).

Two days after electroporation, cells expressing mRuby2 were sorted using FACS, and single cells were plated into 96 well plates in conditioned media. Surviving clones were genotyped by PCR of genomic DNA and screened for phenotype based on centrin-GFP expression.

Primers used for genotyping were: 5'CCCTGCCGACGCAGTGATTGG3' and 5'CAGGGAGTGGCGAGAGCACAC3' for TEDC1 and 5'CTTGCCCGCAAGGAGGGAGAGA3' and 5'GCAGGGCCAGCCAAACAGA3' for TEDC2.

To rescue the mutations, Halo-3xFlag-tagged TEDC1 or APEX-V5-tagged TEDC2 were introduced into the mutant cells using lentiviral transduction as described below.

Lentivirus production and viral transduction

Recombinant lentiviruses were made by cotransfection of HEK293T cells with the respective transfer vectors (TEDC1-HaloTag-3xFlag and TEDC2-V5-APEX2), second-generation lentiviral cassettes (packaging vector psPAX2, pTS3312 and envelope vector pMD2.G, pTS3313) using calcium phosphate-mediated transfection. Briefly, transfection mixture was made with CaCl₂, 2x HBS (50 mM Hepes, 10 mM KCl, 12 mM dextrose, 280 mM NaCl, 1.5 mM Na₂HPO₄×7H₂O, pH 7.05), and plasmids. Cells were treated with 25 μM chloroquine immediately before transfection, then the transfection mixture was added to cells. The medium was changed 5-6 h after transfection, and viral supernatant was harvested after an additional 48 and 72 h. Recipient cells (RPE-1 *TP53*^{-/-}; *TEDC1*^{-/-} and *TP53*^{-/-}; *TEDC2*^{-/-} and *TP53*^{-/-}; *TUBD1*^{-/-} and *TP53*^{-/-}; *TUBE1*^{-/-}) were transduced with viral supernatant and 8 μg/mL Sequabrene. Transduced cells were expanded to 10-cm dishes.

Immunofluorescence

Cells were grown on poly-L-lysine-coated #1.5 glass coverslips (Electron Microscopy Sciences). Cells were fixed with -20°C methanol for 15 min. Coverslips were then washed with PBS for 10 min and blocked with PBS-BT (3% BSA, 0.1% Triton X-100, 0.02% sodium azide in PBS) for 30 min. Coverslips were incubated with primary antibodies diluted in PBS-BT for 1 hr, washed with PBS-BT, incubated with secondary antibodies and DAPI diluted in PBS-BT for 1 hr, then washed again. Samples were mounted using Mowiol (Polysciences) in glycerol containing 1,4-diazobicyclo-[2.2.2]octane (DABCO, Sigma-Aldrich) antifade.

Cell cycle synchronization

For cell cycle analyses in [Fig 1](#), cells were seeded onto coverslips, then synchronized in G0/G1 by serum withdrawal for 24 hr, or in G2 with 10 μM RO-3306 (Adipogen) for 24 hr. Cells were fixed for immunofluorescence and analyzed for centrin/CP110 presence.

For [Fig 4](#) and [5](#), mitotic shakeoff was performed on asynchronously growing cells. One pre-shake was performed to improve synchronization. Cells were fixed for U-ExM and expanded as below.

Expansion microscopy

Ultrastructure Expansion Microscopy (U-ExM)

Cells were grown on poly-D-lysine-coated #1.5 glass coverslips (Electron Microscopy Sciences) and fixed with -20°C methanol for 15 min, then washed with PBS. U-ExM was performed as previously described ([Gambarotto et al., 2019](#)): coverslips were incubated overnight in an acrylamide-formaldehyde anchoring solution (AA/FA; 0.7% formaldehyde, 1% acrylamide in PBS) at 37°C. Gelation was allowed to proceed in monomer solution (19% sodium acrylate, 10% acrylamide, 0.1% bis-acrylamide, 0.5% ammonium persulfate-APS, 0.5% TEMED) for 1 hour at 37°C. Gels were heated in denaturation buffer (200 mM SDS, 200 mM NaCl, 50 mM Tris-HCl pH 9) at 95°C for 1 h.

After denaturation buffer was removed, gels were washed with multiple water rinses and allowed to expand in water at room temperature overnight. Small circles of each expanded gel (~5 mm in diameter) were excised and incubated with primary antibodies diluted in PBS-BT (3% BSA, 0.1% Triton X-100 in PBS) on a nutator at 4°C overnight. The next day, gels were washed three times with PBS-BT buffer and incubated with secondary antibodies and 5 µg/ml DAPI diluted in PBS-BT, protected from light, on a nutator at 4°C overnight.

For **Fig 4**, **Fig 4** – Supp 3 and **Fig 4** – Supp 4 when co-staining with alpha-tubulin, centrioles were fixed with 1.4% formaldehyde and 2% acrylamide for 3 to 5 hours at 37°C. U-ExM was performed as described above. Gels were pre-incubated with anti alpha-tubulin antibody at 4°C overnight prior to staining with other primary antibodies.

Expansion microscopy as per Kong et al

For **Fig 2** – Supp 1C, **D**, expansion microscopy was performed similar to (Kong et al., 2024). Coverslips were incubated in 4% formaldehyde in 1x PBS for 1 hour. The coverslips were then incubated overnight in an acrylamide–formaldehyde anchoring solution (AA/FA; 4% formaldehyde, 30% acrylamide in PBS) at 40°C. Gelation was allowed to proceed in monomer solution (7% sodium acrylate, 20% acrylamide, 0.04% bis-acrylamide, 0.5% ammonium persulfate-APS, 0.5% TEMED in PBS) for 20 min on ice followed by 1 hour at room temperature. Gels were heated in denaturation buffer (200 mM SDS, 200 mM NaCl, 50 mM Tris-HCl pH 9) at 90°C for 1 h. After denaturation buffer was removed, gels were washed with multiple water rinses and allowed to expand in water at room temperature overnight. Small circles of each expanded gel (~5 mm in diameter) were excised and incubated with primary antibodies diluted in PBS-BT (3% BSA, 0.1% Triton X-100 in PBS) on a nutator at 4°C overnight. The next day, gels were washed three times with PBS-BT buffer and incubated with secondary antibodies and 5 µg/ml DAPI diluted in PBS-BT, protected from light, on a nutator at 4°C overnight.

Expansion gel imaging (all protocols)

Immunostained gels were washed once with PBS and at least three times with water, and placed in a glass-bottomed 35 mm plate for imaging. All U-ExM images were acquired as z-stacks collected at 0.27-µm intervals using a confocal Zeiss Axio Observer microscope (Carl Zeiss) with a PlanApoChromat 1.4 NA 63× oil immersion objective, a Yokogawa CSU-W1 (**Fig 2**) or Yokogawa CSU-W1 SoRA head with 2.8x relay (**Fig 4**, **5**) and a Photometrics Prime BSI express CMOS camera. Slidebook software (Intelligent Imaging Innovations, 3i) was used to control the microscope system. Deconvolution was performed with Microvolution (Cupertino, CA) using a calculated point spread function (PSF) for 10 iterations. ImageJ (FIJI) was used for image analysis (Schindelin et al., 2012).

Centriole measurements

For measuring overall centriole width or length, z-stacks of U-ExM images were measured using ImageJ (FIJI) on maximum projections. Only centrioles that were in perfect longitudinal or cross-section were measured. Three measurements were made per centriole and averaged. Measurements were adjusted for gel expansion factor. Statistical analysis was performed with Graphpad Prism.

For measuring protein position as in **Fig 4** – Supp 3 and **Fig 4** – Supp 4, maximum projections of U-ExM images of longitudinally positioned centrioles were measured using ImageJ (FIJI). The coordinates of the proximal-most and distal-most position for each protein were recorded. Three measurements were made per centriole and averaged. The recorded coordinates were used to calculate the positions of the most proximal and most distal signal for each protein, then graphed from shortest to longest centriole.

Transmission electron microscopy

For ultrastructural analysis of centrosomes by TEM, RPE-1 *TP53*^{-/-}; *TEDC1*^{-/-} and RPE-1 *TP53*^{-/-}; *TEDC2*^{-/-} cells were synchronized in G2/M with 10 μ M RO-3306 for 24 hrs. Cells were trypsinized, resuspended in complete media and centrifuged at 800g for 5 min. The pellet was collected in a 14-mL tube and fixed in 2% paraformaldehyde/2.5% glutaraldehyde (Ted Pella Inc., Redding, CA) in 100 mM cacodylate buffer, pH 7.2 for 2 hr at room temperature. Samples were washed in cacodylate buffer and postfixed in 1% osmium tetroxide (Ted Pella Inc.)/1.5% potassium ferricyanide (Sigma, St. Louis, MO) for 1 hr. Samples were then rinsed extensively in dH₂O prior to en bloc staining with 1% aqueous uranyl acetate (Ted Pella Inc.) for 1 hr. Following several rinses in dH₂O, samples were dehydrated in a graded series of ethanol and embedded in Eponate 12 resin (Ted Pella Inc.). Ultrathin sections of 95 nm were cut with a Leica Ultracut UCT ultramicrotome (Leica Microsystems Inc., Bannockburn, IL), stained with uranyl acetate and lead citrate, and viewed on a JEOL 1200 EX transmission electron microscope (JEOL USA Inc., Peabody, MA) equipped with an AMT 8 megapixel digital camera and AMT Image Capture Engine V602 software (Advanced Microscopy Techniques, Woburn, MA).

Symmetrization of TEM images was performed with centrioleJ (<https://www.epfl.ch/labs/gonczy-lab/databases-and-resources/ressources-centriolej/>).

TEDC1 and TEDC2 pulldowns

Cells stably expressing TEDC1-HaloTag-3xFlag or TEDC2-V5-APEX2 were lysed in 50 mM Tris pH7.5, 150 mM NaCl, 1% Triton X-100, 1 mM DTT, Halt protease and phosphatase inhibitor cocktail (ThermoFisher Scientific) for 30 minutes on ice, then cleared by centrifugation at 21,000 g for 20 min. Protein concentration was determined by Pierce BCA Protein Assay -Reducing Agent Compatible (ThermoFisher Scientific). Each cell lysate was incubated with 25 μ L of equilibrated Chromotek Halo-Trap Magnetic Agarose (Proteintech) or Chromotek V5-Trap Magnetic Agarose (Proteintech) for 1 h at 4C on a nutator. Beads were washed using a magnetic separator rack. Elution was performed by adding 80 μ L of 2x SDS loading buffer (100 mM Tris pH 6.8, 4% SDS, 20% glycerol, 100 mM DTT), boiling the beads for 5 min at 95C, then separating the eluate with a magnetic separator rack. Samples were loaded on SDS-PAGE and transferred for Western blotting.

Western blotting

For **Fig 4 - Supp 2**, samples were lysed in 50 mM Tris pH7.5, 150 mM NaCl, 1% Triton X-100, 1 mM DTT, Halt protease and phosphatase inhibitor cocktail (ThermoFisher Scientific) for 30 minutes on ice, then cleared by centrifugation at 21,000 g for 20 min. Protein concentration was determined by Pierce BCA Protein Assay - Reducing Agent Compatible (ThermoFisher Scientific). Equal amounts of protein (20 to 40 μ g) were loaded per lane. For **Fig 3**, samples were loaded after pulldowns.

Proteins were separated by SDS-PAGE and transferred to nitrocellulose (LiCOR Biosciences) in transfer buffer (192 mM Glycine, 25 mM Tris, 20% ethanol). Membranes were blocked with 5% milk in TBST (137 mM NaCl, 25 mM Tris, 2.7 mM KCl, 0.1% Tween-20) at room temp for 1 h, then washed three times with TBST for 5 min each wash. Membranes were incubated with primary antibodies overnight at 4C on a nutator. The next day, membranes were washed three times with TBST for 5 min each wash and incubated with secondary antibodies at room temperature for 2.5 hours. Membranes were washed again with TBST for 5 min each wash and then imaged with the LiCOR Odyssey XF imager and analyzed using Image Studio (LiCOR Biosciences). Each experiment was performed in triplicate.

Antibodies

Primary antibodies used for immunofluorescence and U-ExM and dilutions in PBS-BT: mouse IgG2b anti-acetylated-tubulin, clone 6-11B-1 (1:1000, Sigma-Aldrich Cat# T6793, RRID:AB_477585), rabbit anti-acetyl- α -tubulin (Lys40) (1:100, Cell Signaling Technology Cat# 5335, RRID:AB_10544694), mouse IgG2b anti-centrin3, clone 3e6 (1:1000, Novus Biological, RRID:AB_537701), mouse IgG2a anti-centrin, clone 20H5 (IF 1:200, UExM 1:500, EMD Millipore, RRID:AB_10563501), rat anti-Cep120 (1:1000, gift from Moe Mahjoub (Betleja et al., 2018)), rabbit anti-Cep135 (1:500, Proteintech Cat# 24428-1-AP, RRID:AB_2879543), rabbit anti-Cep295 (1:1000, Sigma-Aldrich Cat# HPA038596, RRID:AB_10672720), rabbit anti-Cep44 (1:100, Proteintech Cat# 24457-1-AP, RRID:AB_2879557), rabbit anti-CENPJ (1:500, Proteintech Cat# 11517-1-AP, RRID:AB_2244605), rabbit anti-CP110 (IF 1:200, UExM 1:2000, Proteintech Cat# 12780-1-AP, RRID:AB_10638480), mouse IgG1 anti-Flag, clone M2 (1:500, Sigma-Aldrich Cat# F1804, RRID:AB_262044), mouse IgG1 anti-gamma-tubulin, clone GTU-88 (IF 1:1000, UExM 1:500, Sigma-Aldrich, RRID:AB_477584), mouse IgG2a anti-PCNA (1:500, BioLegend, RRID:AB_314692), rabbit anti-POC5 (for IF: 1:500, Bethyl Laboratories, RRID:AB_10949152), rabbit anti-POC5 (for U-ExM: 1:500, Thermo Fisher Scientific Cat# A303-341A (also A303-341A-T), RRID:AB_10971172), mouse IgG1 anti-polyglutamylolation, clone GT335 (1:500, AdipoGen Cat# AG-20B-0020, RRID:AB_2490210), rabbit anti-polyglutamate-chain, polyE (1:500, AdipoGen Cat# AG-25B-0030, RRID:AB_2490540), mouse IgG2b anti-SASS6 (1:200, Santa Cruz Cat# sc-81431, RRID:AB_1128357), rabbit anti-STIL (1:500, Abcam Cat# ab89314, RRID:AB_2197878), mouse IgG2a anti-V5 (1:00, Thermo Fisher Scientific Cat# R960-25, RRID:AB_2556564), rabbit anti-WDR90 (1:100, Thermo Fisher Scientific Cat# PA5-61943, RRID:AB_2649628), chicken anti-GFP antibody (Aves Cat #GFP-1020, RRID:AB_10000240).

For immunofluorescence and U-ExM, AlexaFluor conjugated secondary antibodies (Thermo-Fisher) were diluted 1:1000 in PBS-BT. Goat anti-Mouse IgG1, 488 (1:1000, Thermo Fisher Scientific Cat# A-21121, RRID:AB_2535764), Goat anti-Mouse IgG2a, 488 (1:1000, Thermo Fisher Scientific Cat# A-21131, RRID:AB_2535771), Goat anti-Mouse IgG2b, 488 (1:1000, Thermo Fisher Scientific Cat# A-21141, RRID:AB_2535778), Goat anti-rabbit IgG (H+L), 488 (1:1000, Thermo Fisher Scientific Cat# A-11034 (also A11034), RRID:AB_2576217), Goat anti-Mouse IgG1, 568 (1:500, Thermo Fisher Scientific Cat# A-21124, RRID:AB_2535766), Goat anti-Mouse IgG2a, 568 (1:500, Thermo Fisher Scientific Cat# A-21134, RRID:AB_2535773), Goat anti-Mouse IgG2b, 568 (1:500, Thermo Fisher Scientific Cat# A-21144, RRID:AB_2535780), Goat anti-rabbit IgG (H+L), 568 (1:500, Thermo Fisher Scientific Cat# A-11036 (also A11036), RRID:AB_10563566), Goat anti-Mouse IgG3, 594 (1:500, Thermo Fisher Scientific Cat# A-21155, RRID:AB_2535785), Goat anti-rat IgG (H+L), 594 (1:500, Thermo Fisher Scientific Cat# A-11007 (also A11007), RRID:AB_10561522), Goat anti-Mouse IgG1, 647 (1:500, Thermo Fisher Scientific Cat# A-21240, RRID:AB_2535809), Goat anti-Mouse IgG2a, 647 (1:500, Thermo Fisher Scientific Cat# A-21241, RRID:AB_2535810), Goat anti-Mouse IgG2b, 647 (1:500, Thermo Fisher Scientific Cat# A-21242, RRID:AB_2535811), Goat anti-rabbit IgG (H+L), 647 (1:500, Thermo Fisher Scientific Cat# A32733, RRID:AB_2633282), Goat anti-Mouse, Star Red (1:200, Abberior Cat# STRED-1001, RRID:AB_3068620), Goat anti-rabbit, Star Orange (1:200, Abberior Cat #STORANGE-1002, RRID:AB_3068622), Goat anti-chicken, Alexa 488 (Thermo Fisher Scientific Cat# A-11039, RRID:AB_2534096).

Primary antibodies used for Western blotting and dilutions in TBST: rabbit anti TUBD1 (1:1000, Sigma-Aldrich Cat# HPA027090, RRID:AB_1858457), rabbit anti TUBE1 (1:1000, Sigma-Aldrich Cat # HPA032074, RRID:AB_10601216), rabbit anti C14orf80 (1:1000, Sigma-Aldrich Cat # HPA039049, RRID:AB_2676320), rabbit anti C16orf59 (1:1000, Sigma-Aldrich Cat # HPA055389, RRID:AB_2732595), mouse IgG2b anti SASS6 (1:200, Santa Cruz Biotech Cat # sc-81431, RRID:AB_1128357), rabbit anti STIL (1:2000, Abcam Cat# ab89314, RRID:AB_2197878), rabbit anti CENPJ/CPAP (1:1000, Proteintech Cat# 11517-1-AP, RRID:AB_2244605), rabbit anti POC5 (1:1000, Thermo Fisher Scientific Cat# A303-341A (also A303-341A-T), RRID:AB_10971172), mouse IgG2a anti V5 (1:1000, Thermo Fisher Scientific Cat# R960-25, RRID:AB_2556564), mouse IgG1 anti Flag,

clone M2 (1:2000, Sigma-Aldrich Cat# F1804, RRID:AB_262044). Secondary antibodies used for Western blotting: 680 Donkey anti rabbit (H+L) (1:20,000, Thermo Fisher Scientific Cat# A10043, RRID:AB_2534018), 800 Donkey anti rabbit (H+L) (1:20,000, Li-COR Cat# 926-32213, RRID:AB_621848), 680 Donkey anti mouse (H+L) (1:20,000, Thermo Fisher Scientific Cat# A10038, RRID:AB_11180593), 800 Donkey anti mouse (H+L) (1:20,000, Li-COR Cat# 926-32212, RRID:AB_621847).

Acknowledgements

This work was supported by NIH/NIGMS (K99/R00 GM131024 to J.T.W. and R35 GM130286 to T.S.) and Washington University in St. Louis startup funds (to J.T.W). We thank Wendy Beatty of the Washington University Molecular Microbiology Imaging Facility for assistance with transmission electron microscopy, Moe Mahjoub (Washington University School of Medicine) for the gift of the CEP120 and goat anti-rat antibodies, and Meng-Fu Bryan Tsou (Memorial Sloan Kettering Cancer Center) for the gifts of RPE-1 *TP53*^{-/-} and RPE-1 *TP53*^{-/-}; *SASS6*^{-/-} cells. We thank the Stanford cytoskeleton group, WashU centrosome/cilia group, members of the Stearns lab, and David Breslow for helpful discussions. We also thank Hani Zaher's and Joe Jez's labs for hosting the Wang lab during renovations and Larry Galloway for help with Python.

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Reviewer #1 (Public review):

Summary:

The study by Pudlowski et al. investigates how the intricate structure of centrioles is formed by studying the role of a complex formed by delta- and epsilon-tubulin and the TEDC1 and TEDC2 proteins. For this they employ knockout cell lines, EM and ultrastructure expansion microscopy as well as pull-downs. Previous work has indicated a role of delta- and epsilon-tubulin in triplet microtubule formation. Without triplet microtubules centriolar cylinders can still form, but are unstable, resulting in futile rounds of de novo centriole assembly during S phase and disassembly during mitosis. Here the authors show that all four proteins function as a complex and knockout of any of the four proteins results in the same phenotype. They further find that mutant centrioles lack inner scaffold proteins and contain an extended proximal end including markers such as SAS6 and CEP135, suggesting that triplet microtubule formation is linked to limiting proximal end extension and formation of the central region that contains the inner scaffold. Finally, they show that mutant centrioles seem to undergo elongation during early mitosis before disassembly, although it is not clear if this may also be due to prolonged mitotic duration in mutants.

Strengths:

Overall this is a well-performed study, well presented, with conclusions supported by convincing data based on knockout cell lines, rescue experiments, and detailed quantifications.

Weaknesses:

Most weaknesses have been addressed in the revised version. The precise mapping of TED complex proteins to centrioles remains challenging with the available tools but has been addressed through the use of several complementary super-resolution techniques.

<https://doi.org/10.7554/eLife.98704.2.sa3>

Reviewer #2 (Public review):

Summary:

In this article, the authors study the function of TEDC1 and TEDC2, two proteins previously reported to interact with TUBD1 and TUBE1. Previous work by the same group had shown that TUBD1 and TUBE1 are required for centriole assembly and that human cells lacking these proteins form abnormal centrioles that only have singlet microtubules that disintegrate in mitosis. In this new work, the authors demonstrate that TEDC1 and TEDC2 depletion results in the same phenotype with abnormal centrioles that also disintegrate into mitosis. In addition, they were able to localize these proteins to the proximal end of the centriole, a

result not previously achieved with TUBD1 and TUBE1, providing a better understanding of where and when the complex is involved in centriole growth.

Strengths:

The results are very convincing, particularly the phenotype, which is the same as previously observed for TUBD1 and TUBE1. The U-ExM localization is also convincing: despite a signal that's not very homogeneous, it's clear that the complex is in the proximal region of the centriole and procentriole. The phenotype observed in U-ExM on the elongation of the cartwheel is also spectacular and opens the question of the regulation of the size of this structure. The authors also report convincing results on direct interactions between TUBD1, TUBE1, TEDC1, and TEDC2, and an intriguing structural prediction suggesting that TEDC1 and TEDC2 form a heterodimer that interacts with the TUBD1- TUBE1 heterodimer.

Comments on revisions:

I would like to thank the authors for their work and for thoroughly addressing most of my questions. I extend my congratulations to the authors for this excellent and impactful article.

<https://doi.org/10.7554/eLife.98704.2.sa2>

Reviewer #3 (Public review):

Summary:

Human cells deficient in delta-tubulin or epsilon-tubulin form unstable centrioles, which lack triplet microtubules and undergo a futile formation and disintegration cycle. In this study, the authors show that human cells lacking the associated proteins TEDC1 or TEDC2 have these identical phenotypes. They use genetics to knockout TEDC1 or TEDC2 in p53-negative RPE-1 cells and expansion microscopy to structurally characterize mutant centrioles. Biochemical methods and AlphaFold-multimer prediction software are used to investigate interactions between tubulins and TEDC1 and TEDC2.

The study shows that mutant centrioles are built only of A tubules, which elongate and extend their proximal region, fail to incorporate structural components, and finally disintegrate in mitosis. In addition, they demonstrate that delta-tubulin or epsilon-tubulin and TEDC1 and TEDC2 form one complex and that TEDC1 TEDC2 can interact independently of tubulins. Finally, they show that localization of four proteins is mutually dependent.

Strengths:

The results presented here are convincing, exciting, and important, and the manuscript is well-written. The study shows that delta-tubulin, epsilon-tubulin, TEDC1, and TEDC2 function together to build a stable and functional centriole, significantly contributing to the field and our understanding of the centriole assembly process.

Weaknesses:

The ultrastructural characterization of TEDC1 and TEDC2 in centrosomes remains challenging. Nevertheless, it is evident that these proteins occupy growing centrioles and the proximal parts of mother centrioles.

Comments on revisions:

The authors have done a great job extending the original experiments and measurements and answering outstanding questions.

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The study by Pudlowski et al. investigates how the intricate structure of centrioles is formed by studying the role of a complex formed by delta- and epsilon-tubulin and the TEDC1 and TEDC2 proteins. For this, they employ knockout cell lines, EM, and ultrastructure expansion microscopy as well as pull-downs. Previous work has indicated a role of delta- and epsilon-tubulin in triplet microtubule formation. Without triplet microtubules centriolar cylinders can still form, but are unstable, resulting in futile rounds of de novo centriole assembly during S phase and disassembly during mitosis. Here the authors show that all four proteins function as a complex and knockout of any of the four proteins results in the same phenotype. They further find that mutant centrioles lack inner scaffold proteins and contain an extended proximal end including markers such as SAS6 and CEP135, suggesting that triplet microtubule formation is linked to limiting proximal end extension and formation of the central region that contains the inner scaffold. Finally, they show that mutant centrioles seem to undergo elongation during early mitosis before disassembly, although it is not clear if this may also be due to prolonged mitotic duration in mutants.

Strengths:

Overall this is a well-performed study, well presented, with conclusions mostly supported by the data. The use of knockout cell lines and rescue experiments is convincing.

Weaknesses:

In some cases, additional controls and quantification would be needed, in particular regarding cell cycle and centriole elongation stages, to make the data and conclusions more robust.

We thank the reviewer for these comments and have improved our analyses of these as detailed below.

Reviewer #2 (Public Review):

Summary:

In this article, the authors study the function of TEDC1 and TEDC2, two proteins previously reported to interact with TUBD1 and TUBE1. Previous work by the same group had shown that TUBD1 and TUBE1 are required for centriole assembly and that human cells lacking these proteins form abnormal centrioles that only have singlet microtubules that disintegrate in mitosis. In this new work, the authors demonstrate that TEDC1 and TEDC2 depletion results in the same phenotype with abnormal centrioles that also disintegrate into mitosis. In addition, they were able to localize these proteins to the proximal end of the centriole, a result not previously achieved with TUBD1 and TUBE1, providing a better understanding of where and when the complex is involved in centriole growth.

Strengths:

The results are very convincing, particularly the phenotype, which is the same as previously observed for TUBD1 and TUBE1. The U-ExM localization is also convincing:

despite a signal that's not very homogeneous, it's clear that the complex is in the proximal region of the centriole and procentriole. The phenotype observed in U-ExM on the elongation of the cartwheel is also spectacular and opens the question of the regulation of the size of this structure. The authors also report convincing results on direct interactions between TUBD1, TUBE1, TEDC1, and TEDC2, and an intriguing structural prediction suggesting that TEDC1 and TEDC2 form a heterodimer that interacts with the TUBD1- TUBE1 heterodimer.

Weaknesses:

The phenotypes observed in U-ExM on cartwheel elongation merit further quantification, enabling the field to appreciate better what is happening at the level of this structure.

We thank the reviewer for these comments and have improved our analyses of cartwheel elongation as detailed below.

Reviewer #3 (Public Review):

Summary:

Human cells deficient in delta-tubulin or epsilon-tubulin form unstable centrioles, which lack triplet microtubules and undergo a futile formation and disintegration cycle. In this study, the authors show that human cells lacking the associated proteins TEDC1 or TEDC2 have these identical phenotypes. They use genetics to knockout TEDC1 or TEDC2 in p53negative RPE-1 cells and expansion microscopy to structurally characterize mutant centrioles. Biochemical methods and AlphaFold-multimer prediction software are used to investigate interactions between tubulins and TEDC1 and TEDC2.

The study shows that mutant centrioles are built only of A tubules, which elongate and extend their proximal region, fail to incorporate structural components, and finally disintegrate in mitosis. In addition, they demonstrate that delta-tubulin or epsilon-tubulin and TEDC1 and TEDC2 form one complex and that TEDC1 TEDC2 can interact independently of tubulins. Finally, they show that the localization of four proteins is mutually dependent.

Strengths:

The results presented here are mostly convincing, the study is exciting and important, and the manuscript is well-written. The study shows that delta-tubulin, epsilon-tubulin, TEDC1, and TEDC2 function together to build a stable and functional centriole, significantly contributing to the field and our understanding of the centriole assembly process.

Weaknesses:

The ultrastructural characterization of TEDC1 and TEDC2 obtained by U-ExM is inconclusive. Improving the quality of the signals is paramount for this manuscript.

We thank the reviewer for these comments and have improved our imaging of TEDC1 and TEDC2 localization, as detailed below.

Recommendations for the authors:

Reviewing Editor (Recommendations For The Authors):

The reviewers agreed that the conclusions are largely supported by solid evidence, but felt that improving the following aspects would make some of the data more convincing:

(1) The UExM localizations of TEDC1/2 are not very convincing and the reviewers suggest to complement these with alternative super-resolution approaches (e.g. SIM) and/or different labeling techniques such as pre-expansion labeling using STAR red/orange secondaries (also robust for SIM and STED), use of the Halo tag, different tag antibodies, etc

We thank the reviewers for these recommendations and have adapted two of these strategies to improve our imaging of TEDC1 and TEDC2 localization. First, we used an alternative super-resolution approach, a Yokogawa CSU-W1 SoRA confocal scanner (resolution = 120 nm) and imaged cells grown on coverslips (not expanded). We found that TEDC1 and TEDC2 localize to procentrioles and the proximal end of parental centrioles (Fig 2 – Supplementary Figure 1a, b). Second, we used a recently described expansion gel chemistry (Kong et al., Methods Mol Biol 2024) combined with Abberior Star red and orange secondary antibodies. This technique resulted in robust signal at centrosomes and in the cytoplasm and indicated that TEDC1 and TEDC2 localize near the centriole walls of procentrioles and the proximal region of parental centrioles, near CEP44 (Fig 2 – Supplementary Figure 1c, d). These results complement and support our initial observations (Fig 2C, D) and we have edited the text to reflect this (lines 157-163). We also note that these Flag tag and V5 tag primary antibodies are specific and have little background signal in all applications (Fig 2 – Supplementary Fig 1E-J), while other commercially available antibodies against these tags did exhibit non-specific signal.

(2) The cell cycle classifications of centrioles would strongly benefit, apart from a better description, from adding quantifications of average centriole length at a given stage based on tubulin staining (not acTub).

We thank the reviewers for these recommendations. We have added an improved description of our cell cycle analyses (lines 234-237). We have also added new analyses for centriole length as measured by staining with alpha-tubulin (Fig 4 – Supp 3 and Fig 4 – Supp 4). We find that in all mutants, acetylated tubulin elongates along with alpha-tubulin in a similar way as control centrioles.

Reviewer #1 (Recommendations For The Authors):

Specific points:

(1) The introduction is a bit oddly structured. About halfway through it summarizes what is going to be presented in the study, giving the impression that it is about to conclude, but then continues with additional, detailed introduction paragraphs. Overall, the authors may also want to consider making it more concise.

We thank the reviewer for these suggestions and have shortened and restructured the introduction for clarity and conciseness.

(2) The text should explain to the non-expert reader why endogenous proteins are not detected and why exogenously expressed, tagged versions are used. Related to this, the authors state overexpression, but what is this assessment based on? Does expression at the endogenous level also rescue? At least by western blot, these questions should be addressed.

In the text, we have added clarification about why endogenous proteins were not detected for immunofluorescence (lines 149-151). To quantify the overexpression, we have added Western blots of TEDC1 and TEDC2 to Fig 1 – Supplementary Figure 1E,F. We note that endogenous levels of both proteins are very low, and the rescue constructs are overexpressed 20 to 70 fold above endogenous levels.

(3) The figures should clearly indicate when tagged proteins are used and detected. Currently, this info is only found in the legends but should be in the figure panels as well.

We have made these changes to the figure panels in Fig 2, Fig 2 – Supp 1, and Fig 3.

(4) I could not find a description and reference to Figure 2 Supplement 2 and 3.

We have replaced these supplements with new supplementary figures for TEDC1 and TEDC2 localization (Fig 2 – Supp 1).

(5) The multiple bands including unspecific (?) bands should be labeled to guide the reader in the western blots.

We have labeled nonspecific bands in our Western blots with asterisks (Fig 1 – Supp 1, Fig 3)

(6) The alphafold prediction suggests that TUBD1 can bind to the TED complex in the absence of TUBE1 can this be shown? This would be a nice validation of the predicted architecture of the complex. I also missed a bit of a discussion of the predicted architecture. How could it be linked to triplet microtubule formation? Is the latest alphafold version 3 adding anything to this analysis?

In our pulldown experiments, we found that TUBD1 cannot bind to TEDC1 or TEDC2 in the absence of TUBE1 (Fig 3C, D, IB: TUBD1). We performed this experiment with three biological replicates and found the same result. It is possible that TUBD1 and TUBE1 form an intact heterodimer, similar to alpha-tubulin and beta-tubulin, and this will be an exciting area of future research.

We have added new analysis from AlphaFold3 (Fig 3 – Supp 1B). AlphaFold3 predicts a similar structure as AlphaFold Multimer.

We have also added additional discussion about the AlphaFold prediction to the text (lines 220-222, 365-367). Thanks to the reviewer for pointing out this oversight.

(7) I suggest briefly explaining in the text how cells and centrioles at different cell cycle stages were identified. I found some info in the legend of Figure 1, but no info for other figures or in the text. Related to this, how are procentrioles defined in de novo formation? There is no parental centriole to serve as a reference.

We have added a brief explanation of the synchronization and identification in lines 234-237. We have also clarified the text regarding *de novo* centrioles, and now term these “*de novo* centrioles in the first cell cycle after their formation” (lines 271-272).

(8) Related to point 7: using acetylated tubulin as a universal length and width marker seems unreliable since it is a PTM. The authors should use general tubulin staining to estimate centriole dimensions, or at least establish that acetylated tubulin correlates well with the overall tubulin signal in all mutants.

We have added two supplementary data figures (Fig 4 – supp 3 and Fig 4 – supp 4) in which we co-stain control and mutant centrioles with alpha-tubulin. We found that acetylated tubulin marked mutant centrioles well and as alpha-tubulin length increased, acetylated tubulin length also increased.

(9) Presence and absence of various centriolar proteins. These analyses lack a clear reference for the precise centriole elongation stage. This is particularly problematic for proteins that are recruited at specific later stages (such as inner scaffold proteins). The staining should be correlated with centriole length measurements, ideally using general tubulin staining.

As described for point 8, we have added two supplementary data figures in which we costain control and mutant centrioles with alpha-tubulin and found that acetylated tubulin also increases as overall tubulin length increases in all mutants. We note that inner scaffold proteins are absent in all our mutant centrioles at all stages of the cell and centriole cycle, as also previously reported for POC5 in Wang et al., 2017.

Reviewer #2 (Recommendations For The Authors):

Here's a list of points I think could be improved:

- As the authors previously published, the centriole appears to have a smaller internal diameter than mature centrioles. Could the authors measure to see if the phenotype is identical? Is the centriole blocked in the bloom phase (Laporte et al. 2024)?

We have added an additional supplementary figure (Fig 4 – supp 5) to show that mutant centrioles have smaller diameters than mature centrioles, as we previously reported for the delta-tubulin and epsilon-tubulin mutant centrioles by EM. We thank the reviewers for the additional question of the bloom phase. Given the comparatively smaller number of centrioles we analyzed in this paper compared to Laporte et al (50 to 80 centrioles per condition here, versus 800 centrioles in Laporte et al), it is difficult to definitively conclude whether there is a block in bloom phase. This would be an interesting area for future research.

- The images of the centrioles in EM are beautiful. Would it be possible to apply a symmetrisation on it to better see the centriolar structures? For example, is the A-C linker present?

We thank the reviewer for this excellent suggestion. Using centrioleJ, we find that the A-C linker is absent from mutant centrioles. The symmetrized images have been added to Fig 1 – Supplementary Fig 2, and additional discussion has been added to the text (line 143-144, line 368-374).

- How many EM images were taken? Did the centrioles have 100% A-microtubule only or sometimes with B-MT?

For TEM, we focused on centrioles that were positioned to give perfect cross-section images of the centriolar microtubules, and thus did not take images of off-angle or rotated centrioles. Given the difficulty of this experiment (centrioles are small structures within the cell, centrosomes are single-copy organelles, and off-angle centrioles were not imaged), we were lucky to image 3 centrioles that were in perfect cross-section – 2 for *Tedc1*^{-/-} and 1 for *Tedc2*^{-/-}. Our images indicate that these centrioles only have A-tubules (Fig 1 – Supp Fig

- In Figure 2 - it would be preferable to write TEDC2-flag or TEDC1-flag and not TEDC2/1.

We have made this change

- It seems that Figures 2C and D aren't cited, and some of the data in the supplemental data are not described in the main text.

We have replaced these supplements with new supplementary figures for TEDC1 and TEDC2 localization (Fig 2 – Supp 1).

- The signal in U-ExM with the anti-Flag antibody is heterogeneous. Did the authors test several anti-FLAG antibodies in U-ExM?

We tested several anti-Flag and anti-V5 antibodies for our analyses, and chose these because they have little background signal in all applications (Fig 2 – Supplementary Fig 1E-J). Other commercially available antibodies against these tags did exhibit non-specific signal.

- The AlphaFold prediction is difficult to interpret, the authors should provide more views and the PDB file.

We have added 2 additional views of the AlphaFold prediction in Fig 3 – Supp 1A.

- In general, but particularly for Figure 4: the length doesn't seem to be divided by the expansion factor, it is therefore difficult to compare with known EM dimensions. Can the authors correct the scale bars?

We have corrected the scale bars for all figures to account for the expansion factor.

- Concerning Gamma-tubulin that is "recruited to the lumen of centrioles by the inner scaffold, had localization defects in mutant centrioles. However, we were unable to reliably detect gamma-tubulin within the lumen of control or de novo-formed centrioles in S or G2-phase (Figure 4 - Supplement 1E), and thus were unable to test this hypothesis". In Laporte et al 2024, Gamma-tubulin arrives later than the inner scaffold and only on mature centrioles, so this result appears to be in line with previous observation. However, the authors should be able to detect a proximal signal under the microtubules of the procentriole, is this the case?

We agree that this is an exciting question. However, in our expansion microscopy staining, we frequently observe that gamma-tubulin surrounds centrioles, corresponding to its role in the pericentriolar material (PCM). In our hands, we find it difficult to distinguish between centriolar gamma-tubulin at the base of the A-tubule from gamma-tubulin within the PCM.

- In the signal elongation of SAS-6, STIL, CEP135, CPAP, and CEP44, would it be possible to quantify the length of these signals (with dimensions divided by the expansion factor for comparison with known TEM distances)?

We have quantified the lengths of SAS-6 and CEP135 in new Fig 4 – Supp 3 and Fig 4 – Supp 4.

- The authors observe that centrin is present, but only as a SFI1 dot-like localization (which is another protein that would be interesting to look at), and not an inner scaffold localization. Can the authors elaborate? These results suggest that the distal part is correctly formed with only a microtubule singlet.

We agree with the reviewer's interpretation that the centriole distal tip is likely correctly formed with only singlet microtubules, as both distal centrin and CP110 are present. We have

added this point to the discussion (line 415).

-The authors observe that CPAP is elongated, but CPAP has two locations, proximal and distal. Is it distal or proximal elongation? Is the proximal signal of CPAP longer than that of CEP44 in the mutants? The authors discuss that the elongation could come from overexpression of CPAP, but here it seems that the centriole is not overlong, just the structures around the cartwheel.

We thank the reviewer for this point. It is difficult for us to conclude whether the proximal or distal region is extended in the mutants, as our mutant centrioles lack a visible separation between these two regions. It would be interesting to probe this question in the future by testing whether subdomains of CPAP may be differentially regulated in our mutants.

Reviewer #3 (Recommendations For The Authors):

It isn't apparent to me what was counted in Figure 1C. Were all centrioles (mother centrioles and procentrioles) counted? Where is the 40% in control cells coming from? Can this set of data be presented differently?

We apologize for the confusion. In this figure, all centrioles were counted. We have updated the figure legend for clarity. We performed this analysis in a similar way as in Wang et al., 2017 to better compare phenotypes.

Figure 2C. and the text lines 182-187: The ultrastructural characterization of TEDC1 and TEDC2 suffers from the low quality of the TEDC1 and TEDC2 signals obtained postexpansion. In comparison with robust low-resolution immunosignal, it appears that most of the signal cannot be recovered after expansion. Another sub-resolution imaging method to re-analyze TEDC1 and TEDC2 localization would be essential. The same concern applies to Figures 2 - Supplement 2 and 3. Also, Figure 2 - Supplement 2 and Supplement 3 do not seem to be cited.

We thank the reviewer for these recommendations. As also mentioned above, we used an alternative super-resolution approach, a Yokogawa CSU-W1 SoRA confocal scanner (resolution = 120 nm), and found that TEDC1 and TEDC2 localize to procentrioles and the proximal end of parental centrioles (Fig 2 – Supplementary Figure 1a, b). Second, we used a recently described expansion gel chemistry (Kong et al., Methods Mol Biol 2024) combined with Abberior Star red and orange secondary antibodies. This technique resulted in robust signal at centrosomes and in the cytoplasm and indicated that TEDC1 and TEDC2 localize near the centriole walls of procentrioles and the proximal region of parental centrioles, near CEP44 (Fig 2 – Supplementary Figure 1c, d). These stainings complement and support our initial observations (Fig 2C, D) and we have edited the text to reflect this (lines 157-163). We have also removed the supplementary figures that were uncited in the text.

TUBD1 and TUBE1 form a dimer and TEDC2 and TEDC1 can interact. Any speculation as to why TEDC2 does not pull down both TUBE1 and TUBD1?

We apologize for the confusion. TEDC2 does pull down both TUBE1 and TUBD1 (Fig 3D, pull-down, second column, Tedc2-V5-APEX2 rescuing the *Tedc2*^{-/-} cells pulls down TUBD1, TUBE1, and TEDC1).

Figure 4A and B. The authors use acetylated tubulin to determine the length of procentrioles in the S and G2 phases. However, procentrioles are not acetylated on their distal ends in these cell phase phases (as the authors also mention further in the text). Why has alpha tubulin not been used since it works well in U-ExM? The average size of

the control, G2 procentrioles, seems too small in Figure 4A and not consistent with other imaging data (for instance, in Figure 4 - Supplement 1 C, Cp110, and CPAP staining). There is no statistical analysis in F4A.

We have added two supplementary data figures (Fig 4 – supp 3 and Fig 4 – supp 4) in which we co-stain control and mutant centrioles with alpha-tubulin. We found that acetylated tubulin correlates well with overall tubulin signal in all mutants. We have added statistical analysis to the figure legend of Fig 4A.

Lines 260 - 262: "These results indicate that centrioles with singlet microtubules can elongate to the same length as controls, and therefore that triplet microtubules are not essential for regulating centriole length." It is hard to agree with this statement. Mutant procentrioles show aberrantly elongated proximal signals of several tested proteins. In addition, in lines 326 - 328, the authors state that "Together, these results indicate that centrioles lacking compound microtubules are unable to properly regulate the length of the proximal end."

We thank the reviewer and have clarified the statement to state that these results indicate that centrioles with singlet microtubules can elongate to the same overall length as control centrioles in G2 phase.

Line 353: The authors suggest that elongated procentriole structure in mitosis may represent intermediates in centriole disassembly. Another interpretation, more in line with the EM data from Wang et al., 2017, would be that these mutant procentrioles first additionally elongate before they disassemble in late mitosis. The aberrant intermediate structure concept would need further exploration. For instance, anti-alpha/beta-tubulin antibodies could be used to investigate centriole microtubules.

We apologize for the confusion and have edited this section for clarity (lines 341-343): “We conclude that in our mutant cells, centrioles elongate in early mitosis to form an aberrant intermediate structure, followed by fragmentation in late mitosis.”

References need to be included in lines 122, 277, 279.

We have added these references

Line 281: Add references PMID: 30559430 and PMID: 32526902.

We have added these references (lines 265-266).

Line 289: "Moreover, our results suggest that centriole glutamylation is a multistep process, in which long glutamate side chains are added later during centriole maturation." This does not seem like an original observation. For instance, see PMID: 32526902.

We have added this reference (lines 273-274).

<https://doi.org/10.7554/eLife.98704.2.sa0>